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ECONOMICS • UPPER SIXTH | UPPER SIXTH — ECONOMICS

CHAPTER 1

NATIONAL INCOME ACCOUNTING

SUGGESTED TIME: 7 PERIODS

IN THIS CHAPTER

This chapter introduces national income accounting: the meaning of key aggregates (GDP, GNP, NNP, national income), the three approaches to measuring them, and the practical calculation of national income from Cameroonian-style data. It also examines the problems of measurement, their solutions, and the importance of national income statistics for policy and the GCE examination.

- By the end of this chapter I can define national income and distinguish between GDP, GNP, NNP (at market prices and at factor cost), personal income and disposable income.
- By the end of this chapter I can explain the circular flow of income and why the output, income and expenditure approaches yield the same total.
- By the end of this chapter I can calculate national income using the expenditure approach from a given set of data.
- By the end of this chapter I can calculate national income using the income approach, avoiding double counting of transfer payments.
- By the end of this chapter I can calculate national income using the output (value added) approach, avoiding double counting of intermediate goods.
- By the end of this chapter I can convert GDP at market prices into GNP at factor cost and into national income using net property income from abroad, indirect taxes, subsidies and depreciation.

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CONTENTS

1. National income: concepts and key aggregates
2. The expenditure approach (Practical Work 12)
3. The income and output approaches (Practical Work 13)
4. Problems of measuring national income and solutions
5. Importance and limitations of national income statistics
6. Integration activity
7. Methods and worked examples
8. Exercises
9. Solutions to the exercises
10. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can define national income and distinguish between GDP, GNP, NNP (at market prices and at factor cost), personal income and disposable income.
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- By the end of this chapter I can convert GDP at market prices into GNP at factor cost and into national income using net property income from abroad, indirect taxes, subsidies and depreciation.
- By the end of this chapter I can identify the problems of measuring national income in Cameroon and suggest practical solutions.
- By the end of this chapter I can explain the importance and limitations of national income statistics for comparing living standards over time and between countries.

PREREQUISITES

- The circular flow of income between households and firms (Lower Sixth)
- Basic macroeconomic objectives: growth, employment, price stability (Lower Sixth)
- Distinction between goods and services, and between consumption and investment
- Elementary percentage and table calculations in economics
- The concept of factors of production and their rewards (rent, wages, interest, profit)

1 NATIONAL INCOME: CONCEPTS AND KEY AGGREGATES

In a recent year, Cameroon's Ministry of Finance announced that the country's GDP had grown, yet a trader at Mokolo market in Yaoundé complained that her daily sales had fallen. How can the nation be "richer" while ordinary citizens feel poorer? Who actually counts the value of all the yams, phone credits, road works and banking services produced in Cameroon each year, and how? This chapter takes you inside the statistical workshop of the nation to show how national income is defined, how it is measured, and why its figures must be read with care.

1.1 The meaning of national income

Imagine a single household in Buea. At the end of the month, the family can say exactly how much income it received: salaries, the profit from a small provision store, perhaps remittances from a relative in Douala. A nation, however, has millions of households and firms. To know whether the country is becoming richer or poorer, economists must add up, in a systematic way, the value of everything produced and every income earned over a period. That grand total is what we call **national income**.

DEFINITION **National income**

National income is the total money value of all final goods and services produced by the residents of a country during a period of one year; equivalently, it is the total of all incomes (wages, rent, interest and profit) earned by the factors of production owned by the country's residents during that year.

Notice the two faces of the same coin in this definition: national income can be seen as a flow of output (goods and services produced) or as a flow of income (rewards to factors of production). This double identity is the foundation of national income accounting, and it is best understood through the **circular flow of income**.

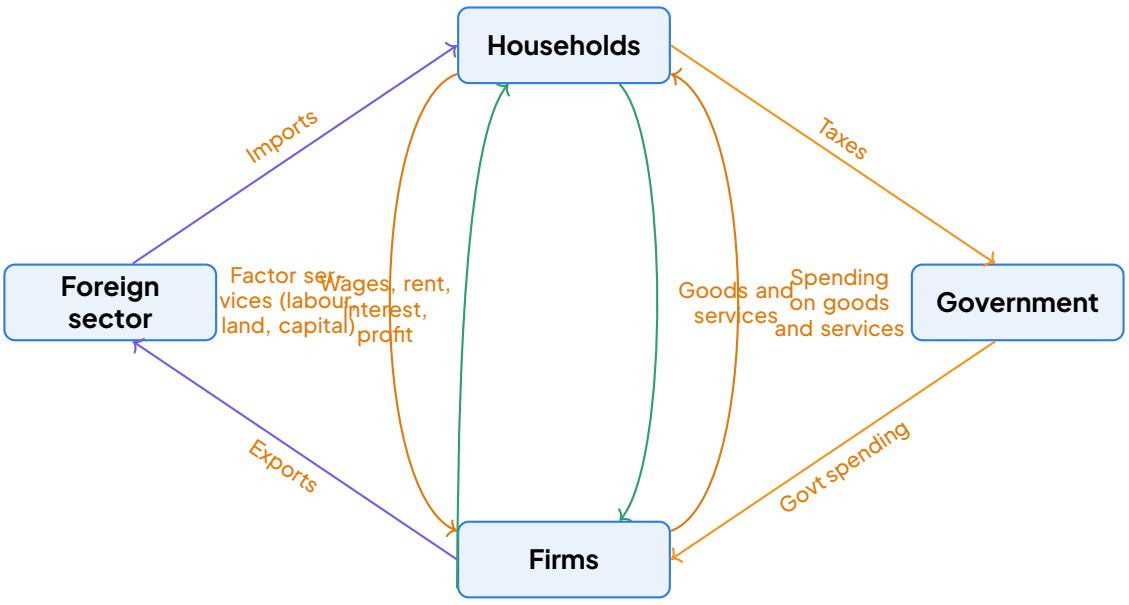
Two words in the definition deserve emphasis. First, only **final** goods and services are counted: a final good is one sold to its ultimate user, whereas an intermediate good (for example, cocoa beans sold to a processing factory) is used up in producing something else. Counting both the beans and the chocolate would count the same value twice. Second, the period is a flow of one year: national income is a flow concept, not a stock of wealth.

1.2 The circular flow of income: why output = income = expenditure

Consider a simplified economy with only households and firms. Households supply labour, land and capital to firms and receive wages, rent, interest and profit in return. Firms use these factors to produce goods and services, which households buy with their income. Every franc spent by a buyer becomes a franc of revenue for a seller, which in turn becomes income for a factor owner. Hence the value of output equals the income generated, which equals the expenditure on that output.

The real economy adds two more actors: the government (which taxes and spends) and the

foreign sector (which buys our exports and sells us imports). These create leakages (or withdrawals: savings, taxes, imports) and injections (investment, government spending, exports) into the flow.



The circular flow of income between households, firms, government and the foreign sector. Output, income and expenditure are three views of the same circular flow.

KEY POINTS **The fundamental identity**

Because every expenditure is somebody's income, and every income arises from producing output, national income can be measured in three equivalent ways:

$$\text{Output} = \text{Income} = \text{Expenditure}.$$

This identity justifies the three approaches (output, income, expenditure) that will be used in the practical works of this chapter.

1.3 Gross Domestic Product (GDP)

The most famous aggregate is GDP. Its key feature is the territorial criterion: it counts production that takes place **inside the country's borders**, whoever owns the factors of production.

DEFINITION **Gross Domestic Product**

Gross Domestic Product (GDP) is the total market value of all final goods and services produced within the geographical territory of a country during one year, whether the pro-

ducing factors are owned by residents or by foreigners.

Thus the output of a foreign-owned cocoa-processing factory in Douala counts in Cameroon's GDP, because the production occurs on Cameroonian territory.

GDP at market prices vs GDP at factor cost. When goods are valued at the prices actually paid in shops, those prices include indirect taxes (such as VAT) and are reduced by subsidies. The income actually received by factor owners excludes indirect taxes and includes subsidies. Hence:

$$\text{GDP at factor cost} = \text{GDP at market prices} - \text{indirect taxes} + \text{subsidies}.$$

METHOD **From market prices to factor cost**

1. Start with the aggregate valued at **market prices** (prices paid by buyers).
2. Subtract indirect taxes (VAT, excise duties): they go to the government, not to factor owners.
3. Add subsidies: they are paid by the government to producers, so they raise factor incomes above what market prices alone would give.
4. The result is the aggregate at **factor cost**, i.e. measured by the incomes earned by factors of production.

1.4 Gross National Product (GNP) and net property income from abroad

GDP measures where production occurs; GNP measures who owns the producing factors. The criterion is now nationality (residence) rather than territory.

DEFINITION **Gross National Product and NPIA**

Gross National Product (GNP) is the total market value of all final goods and services produced in one year by the factors of production owned by the residents of a country, whether those factors are located at home or abroad. **Net property income from abroad (NPIA)** is the difference between property income (profits, dividends, interest, rent) earned by residents from abroad and property income earned by foreigners within the country:

$$\text{GNP} = \text{GDP} + \text{NPIA}.$$

For Cameroon, NPIA is typically negative: the profits repatriated by foreign companies operating in the oil, timber and telecommunications sectors exceed the income that Cameroonian residents earn abroad. Cameroon's GNP is therefore usually smaller than its GDP.

WARNING **Common mistake: confusing GDP and GNP**

A frequent GCE error is to treat GDP and GNP as interchangeable. Remember the criterion:

- GDP = territory: production inside Cameroon, even by foreign firms.
- GNP = nationals: production by Cameroonian residents, even abroad.

The profits of a foreign firm operating in Yaoundé count in Cameroon's GDP but are then subtracted (through NPIA) when computing GNP. Conversely, the earnings of a Cameroonian consultant working in Gabon count in Cameroon's GNP, not its GDP.

1.5 Net National Product (NNP): national income proper

Capital wears out. Machines in a Douala brewery depreciate, lorries on the Douala–Yaoundé road need replacing, buildings age. Part of each year's gross output must be set aside simply to replace worn-out capital; only the remainder is genuinely available for consumption or net addition to wealth. This allowance is called **depreciation** or capital consumption.

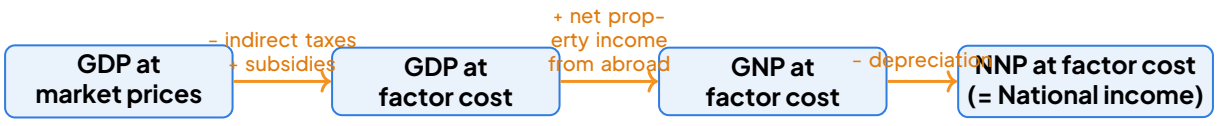
DEFINITION **Net National Product**

Net National Product (NNP) is GNP minus depreciation (capital consumption):

$$\text{NNP} = \text{GNP} - \text{depreciation}.$$

NNP measured at factor cost is what economists call **national income** in the strict sense: the total income actually earned by the nation's factors of production.

The chain of conversions is summarised in the following flowchart.



From GDP at market prices to national income (NNP at factor cost). Each arrow applies one adjustment.

WORKED EXAMPLE 1 **Worked example: from GDP to national income**

The National Institute of Statistics of a hypothetical CEMAC country reports the following for the year (billions of FCFA): GDP at market prices = 25 000; indirect taxes = 2 800; sub-

sidies = 500; net property income from abroad = -600; depreciation = 2 100. Calculate (a) GDP at factor cost, (b) GNP at factor cost, (c) national income.

SOLUTION

Solution.

$$\text{(a)}\quad GDP_{fc} = GDP_{mp} - \text{indirect taxes} + \text{subsidies}$$
$$= 25\,000 - 2\,800 + 500 = 22\,700 \text{ billion FCFA.}$$
$$\text{(b)}\quad GNP_{fc} = GDP_{fc} + \text{NPIA}$$
$$= 22\,700 + (-600) = 22\,100 \text{ billion FCFA.}$$
$$\text{(c)}\quad \text{National income} = NNP_{fc} = GNP_{fc} - \text{depreciation}$$
$$= 22\,100 - 2\,100 = 20\,000 \text{ billion FCFA.}$$

Note how the negative NPIA (profits repatriated by foreign firms) reduces GNP below GDP, exactly as is typical for Cameroon.

1.6 Personal income and disposable personal income

National income is earned by factors of production, but it is not exactly what households can spend. Some income never reaches households (undistributed company profits, corporate taxes, social security contributions), while households receive some income that was not earned by current production: **transfer payments** such as pensions, family allowances and scholarships.

DEFINITION **Personal income and disposable personal income**

Personal income is the total income actually received by households before paying direct taxes:

Personal income = National income−undistributed profits−corporate taxes+transfer payments.

Disposable personal income is personal income minus direct taxes (income tax, property tax). It is what households can actually spend or save.

WARNING **Common mistake: counting transfer payments as national income**

Pensions, scholarships and family allowances are transfers: they are paid out of taxes raised from current production, and no new good or service is produced in exchange. They must not be added when computing national income, otherwise the same income would be counted twice (once when earned by the taxpayer, once when received by the

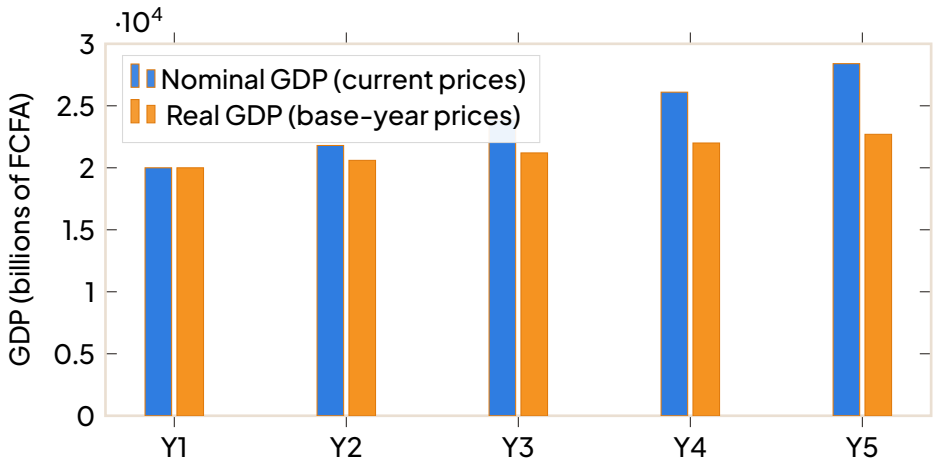
pensioner). Transfers enter only the move from national income to personal income.

1.7 Nominal vs real national income and per capita income

If the price of a bag of garri doubles, money GDP may double without a single extra bag being produced. To separate genuine growth from pure inflation, economists distinguish:

DEFINITION **Nominal and real national income**

Nominal national income (or money national income) is output valued at current prices of the year in question. **Real national income** is output valued at the constant prices of a chosen base year, so that changes reflect only changes in the volume of production.



Nominal versus real GDP of a hypothetical economy over five years. Nominal GDP grows faster because it includes rising prices; real GDP isolates the true growth of output.

Finally, to compare living standards across countries or over time, total income must be related to population:

DEFINITION **Per capita income**

Per capita income is national income (or GDP) divided by the total population:

$$\text{Per capita income} = \frac{\text{National income}}{\text{Population}}.$$

In Cameroon, these figures are compiled and published by the **National Institute of Statistics (INS)**, which conducts household surveys, enterprise censuses and price collections to build

the national accounts. This explains our opening puzzle: GDP can grow while a Mokolo trader feels poorer, because (i) growth may be concentrated in sectors such as oil or construction that do not reach her stall, (ii) population growth may absorb much of the increase, leaving per capita income nearly flat, and (iii) inflation may mean nominal growth exceeds real growth.

DID YOU KNOW? **Did you know?**

Cameroon's national accounts are produced within the CEMAC statistical framework and use the CFA franc as the unit of account. Because a large share of Cameroonian activity takes place in the informal sector (roadside trade, smallholder food farming, artisanal transport), the INS must estimate much of GDP indirectly — one reason why official figures are revised from time to time and why they understate true economic activity.

EXERCISE 1 **Practice**

The following data (billions of FCFA) relate to a hypothetical economy in one year: GDP at market prices = 30 000; indirect taxes = 3 200; subsidies = 800; income earned by residents abroad = 900; income earned by foreigners domestically = 1 500; depreciation = 2 500; population = 25 million.

1. Calculate GDP at factor cost.
2. Calculate net property income from abroad and GNP at factor cost.
3. Calculate national income (NNP at factor cost).
4. Calculate per capita national income in FCFA.
5. Explain why transfer payments were not needed in any of the above calculations.

SOLUTION **Exercise 1**

1. GDP_{fc} = 30 000 − 3 200 + 800 = 27 600 billion FCFA.
2. NPIA = 900 − 1 500 = −600 billion FCFA. GNP_{fc} = 27 600 − 600 = 27 000 billion FCFA.
3. NNP_{fc} = 27 000 − 2 500 = 24 500 billion FCFA.
4. Per capita income = $\frac{24\,500 \text{ billion FCFA}}{25 \text{ million}}$ = 980 000 FCFA per person.
5. Transfer payments merely redistribute existing income from taxpayers to recipients; they do not correspond to any new production, so including them would double-count income already measured in the output of the economy.

KEY POINTS **Exam tip: "distinguish between" questions**

GCE Advanced Level questions frequently ask candidates to distinguish between two aggregates (GDP and GNP; market prices and factor cost; nominal and real income; national income and disposable income). Always state explicitly the criterion of difference: territory versus nationality for GDP/GNP; inclusion or exclusion of indirect taxes and subsidies for market prices/factor cost; current versus constant prices for nominal/real; earned income versus income actually available to households for national/disposable income. One clear criterion, plus a short Cameroonian illustration, earns full marks.

2 THE EXPENDITURE APPROACH (PRACTICAL WORK 12)

In the previous section we defined national income and the key aggregates (GDP, GNP, NNP) and we saw that the circular flow of income guarantees that the value of production, the value of incomes and the value of expenditure are three ways of looking at the same economic activity. We now exploit one side of that identity: expenditure. The expenditure approach is the most commonly used method in practice — including by Cameroon's National Institute of Statistics — because spending data are relatively easy to observe through surveys, customs records and government accounts.

2.1 The principle: summing all final expenditure

Imagine you could stand at every shop, market stall, building site, ministry and port in Cameroon during one year and record every franc spent on **newly produced, final** goods and services. Add all those francs together and you obtain the value of everything the economy produced. That is the whole intuition of the expenditure approach: every good produced is, sooner or later, bought by someone, so total spending on final output must equal total output.

DEFINITION **The expenditure approach**

The **expenditure approach** measures national income (starting from GDP) by summing all expenditure on final goods and services produced within the country during a given period (usually one year). Spending on intermediate goods, on second-hand goods and on purely financial transactions is excluded.

Who does the spending? Every buyer in the economy falls into one of four groups, and this gives us the four components of the approach:

1. **Households** buy consumer goods and services: this is **consumption** (C) — food, clothing, transport, school fees, airtime.
2. **Firms** buy capital goods and build up stocks: this is **gross private domestic investment** (I) — machinery, factories, new buildings, changes in inventories.

3. **The State** spends on public services and public investment: this is **government expenditure** (G) — salaries of civil servants, roads, hospitals.
4. **Foreigners** buy our goods (exports, X), while we buy foreign goods (imports, M). The net contribution of the rest of the world is **net exports** ($X - M$).

2.2 The fundamental identity

PROPERTY **Expenditure identity**

$$\text{GDP}_{mp} = C + I + G + (X - M)$$

where GDP is measured at market prices. This is an identity: it is true by definition of the four components. It is not a theorem to be proved, but its justification (why each component appears, and why imports are subtracted) is frequently demanded in the examination.

Why are imports subtracted? This is the point students most often misunderstand. Consumption C, investment I and government spending G are measured without distinguishing where the goods were produced. When a household in Douala buys an imported television, that spending is already inside C — but the television was not produced in Cameroon, so it must not count in Cameroon's GDP. Imports M therefore appear twice in logic: once hidden inside C, I and G (with a plus sign), and once explicitly in $(X - M)$ (with a minus sign). The two cancel, leaving only spending on domestically produced output. Exports, conversely, are produced at home but bought by foreigners, so they are not in C, I or G and must be added.

WARNING **Common mistake: adding imports**

Many candidates write GDP $= C+I+G+X+M$. This is wrong and costs several marks. Imports are subtracted, because they are foreign production already included in C, I and G. A useful memory device: GDP measures what we produce; imports are what they produce — so imports must leave the sum.

2.3 Final goods versus intermediate goods

Only expenditure on **final** goods and services is counted. A **final good** is one bought by its ultimate user for consumption or investment; an **intermediate good** is one bought by a firm to be used up or transformed in the production of another good.

Consider a baguette sold in Yaoundé. The flour bought by the bakery is an intermediate good: its value is already embodied in the price of the bread. If we counted both the flour and the bread, we would count the flour's value twice — this is **double counting**. The expenditure approach avoids it automatically, because only the final purchase (the household buying the bread) enters C.

WARNING **Common mistake: counting intermediate expenditure**

Flour bought by a bakery, cocoa beans bought by a chocolate factory, cement bought by a construction firm: none of these enter GDP directly. Only the final sale counts. In an examination, when you are given a list of transactions, always ask: "Is this buyer the final user, or a firm that will transform the good?"

Note also that **second-hand sales** are excluded: when you sell your used motorcycle, no new production has taken place this year; the motorcycle was counted in GDP in the year it was produced. Only the dealer's commission (a newly produced service) counts.

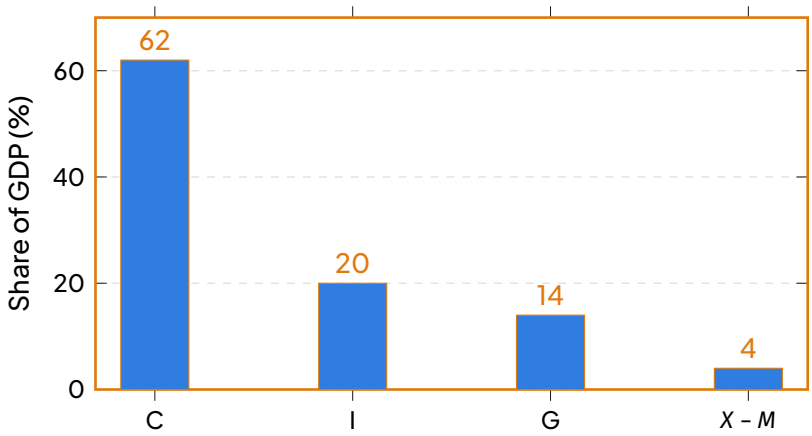
2.4 From GDP at market prices to national income: the adjustments

The expenditure identity gives us GDP_{mp}. But the syllabus requires national income, which is a narrower concept: the income actually earned by a country's residents from current production, measured at factor cost. Four adjustments are needed.

METHOD **From GDP(mp) to national income**

1. List all expenditure items given in the question.
2. Classify each item into C, I, G, X or M (reject intermediate goods, second-hand sales, transfer payments).
3. Sum: $GDP_{mp} = C + I + G + (X - M)$.
4. Adjust step by step:
   - Subtract **indirect taxes** and add **subsidies** to move from market prices to factor cost: $GDP_{fc} = GDP_{mp}$ − indirect taxes + subsidies.
   - Add **net property income from abroad** (NPIA = income earned by residents abroad minus income earned by foreigners at home) to move from domestic to national: $GNP_{fc} = GDP_{fc} + NPIA$.
   - Subtract **depreciation** (capital consumption) to move from gross to net: $NNP_{fc} = GNP_{fc}$ − depreciation. This NNP_{fc} is national income.

Why these adjustments? Indirect taxes (such as VAT) raise market prices above what factors of production actually receive, while subsidies do the opposite; national income measures factor earnings, so taxes must come out and subsidies go in. Depreciation is the part of gross investment that merely replaces worn-out capital: it is a cost of production, not income available to anyone, so it is deducted. NPIA corrects for the fact that some domestic production rewards foreign-owned factors (and vice versa) — for instance, profits repatriated by foreign companies operating in Cameroon reduce Cameroon's national income relative to its GDP.



Hypothetical shares of the four expenditure components in Cameroonian GDP: consumption dominates, while net exports contribute only a small positive amount.

The bar chart illustrates a typical pattern for an economy like Cameroon's: household consumption is by far the largest component, followed by investment and government spending, with net exports small (and in many years negative, when imports exceed exports).

2.5 Standard layout of the calculation

In Practical Work, presentation matters as much as the arithmetic. The examiner expects a clear vertical statement in which every adjustment appears on its own line.

Expenditure approach statement (billions of FCFA)	
Consumption (C)	xxx
+ Gross private domestic investment (I)	xxx
+ Government expenditure (G)	xxx
+ Exports (X)	xxx
- Imports (M)	(xxx)
= GDP at market prices	**xxx**
- Indirect taxes + Subsidies	(xxx)
= GDP at factor cost	**xxx**
+ Net property income from abroad	xxx
= GNP at factor cost	**xxx**
- Depreciation	(xxx)
= NNP at factor cost = NATIONAL INCOME	**xxx**

Template for the expenditure-approach calculation. Reproduce this layout in the examination, one adjustment per line.

2.6 Guided worked example with Cameroonian-style data

WORKED EXAMPLE 2 **Expenditure approach: full calculation**

The following data (in billions of FCFA) describe a hypothetical economy for one year: household consumption 6 200; government current expenditure 1 400; gross fixed capital formation 1 900; increase in stocks 100; exports of goods and services 2 300; imports of goods and services 2 600; indirect taxes 850; subsidies 150; net property income from abroad -250; depreciation 700. Calculate (a) GDP at market prices, (b) GDP at factor cost, (c) GNP at factor cost, (d) national income.

SOLUTION

Step 1 — Classify. Consumption C = 6 200; government G = 1 400; investment I = 1 900 + 100 = 2 000 (gross fixed capital formation plus the change in stocks); exports X = 2 300; imports M = 2 600.
Step 2 — GDP at market prices.

$$\begin{aligned} \text{GDP}_{mp} &= C + I + G + (X - M) \\ &= 6\,200 + 2\,000 + 1\,400 + (2\,300 - 2\,600) \\ &= 9\,600 - 300 = 9\,300 \text{ billion FCFA.} \end{aligned}$$

Note the negative net exports: imports exceed exports by 300, which reduces GDP.
Step 3 — GDP at factor cost.

$$\text{GDP}_{fc} = \text{GDP}_{mp} - \text{indirect taxes} + \text{subsidies} = 9\,300 - 850 + 150 = 8\,600 \text{ billion FCFA.}$$

Step 4 — GNP at factor cost.

$$\text{GNP}_{fc} = \text{GDP}_{fc} + \text{NPIA} = 8\,600 + (-250) = 8\,350 \text{ billion FCFA.}$$

The negative NPIA means foreigners earn more from this economy than its residents earn abroad.
Step 5 — National income (NNP at factor cost).

$$\text{NNP}_{fc} = \text{GNP}_{fc} - \text{depreciation} = 8\,350 - 700 = 7\,650 \text{ billion FCFA.}$$

Answers: (a) 9 300; (b) 8 600; (c) 8 350; (d) 7 650 billion FCFA.

2.7 Practice pitfalls and examination technique

Three errors recur every year in scripts, and all are easily avoided.

First, **including second-hand sales**: the resale of a used car or a second-hand "okrika" bale is not current production. Exclude the sale value entirely; count only any dealer's or agent's fee, which is a new service.

Second, **double counting**: never add the value of intermediate goods (flour for the bakery, alu-

minium for the saucepan factory) on top of final goods. If a question gives you both, keep only the final sale.

Third, **omitting or mishandling net exports**: some candidates forget $(X - M)$ altogether; others add M. Always compute $X - M$ explicitly, and do not panic if it is negative — a trade deficit simply reduces GDP.

WARNING **Exam tip: presentation earns marks**

Always state the unit ("billions of FCFA") in every answer line, and show every adjustment on its own line — indirect taxes, subsidies, NPIA, depreciation — even if its value is zero or negative. GCE mark schemes allocate marks line by line; a candidate who jumps straight from GDP_{mp} to national income loses the method marks even with a correct final figure.

EXERCISE 2 **Practice: expenditure approach**

From the following hypothetical data (billions of FCFA), calculate GDP at market prices and national income: consumption 4 500; government expenditure 1 100; gross investment 1 300; exports 1 800; imports 2 100; indirect taxes 600; subsidies 100; NPIA -150; depreciation 500.

SOLUTION **Practice: expenditure approach**

GDP_{mp} = 4 500 + 1 300 + 1 100 + (1 800 - 2 100) = 6 600 billion FCFA. Then GDP_{fc} = 6 600 - 600 + 100 = 6 100; GNP_{fc} = 6 100 - 150 = 5 950; national income = NNP_{fc} = 5 950 - 500 = 5 450 billion FCFA.

KEY POINTS **Key points**

- The expenditure approach sums spending on final, domestically produced goods and services: $GDP_{mp} = C + I + G + (X - M)$.
- Imports are subtracted because they are foreign output already hidden inside C, I and G.
- Intermediate goods and second-hand sales are excluded to avoid double counting.
- National income = GDP_{mp} - indirect taxes + subsidies + NPIA - depreciation (i.e. NNP at factor cost).
- Present the calculation line by line, always with units.

Having mastered the expenditure side of the circular flow, we turn in the next section to the two other routes to the same total: the income approach and the output (value added) approach.

3 THE INCOME AND OUTPUT APPROACHES (PRACTICAL WORK 13)

Having mastered the expenditure approach in Practical Work 12, we now turn to the two remaining methods of measuring national income. The **income approach** adds up all factor incomes earned by residents, while the **output (value-added) approach** sums the value added at each stage of production. Both must yield the same aggregate as the expenditure approach — a fundamental consistency check that examiners love to test.

3.1 The income approach

DEFINITION **Income approach**

The income approach measures Gross Domestic Product (GDP) at market prices by summing all factor incomes generated in the production of goods and services within the domestic territory during a given period. The main components are:

- **Wages and salaries** (compensation of employees)
- **Rent** (income from land and property)
- **Interest** (income from lending capital)
- **Profits** (including dividends and undistributed profits)
- **Income of the self-employed** (mixed income)

To obtain GDP at market prices we add **depreciation** (consumption of fixed capital) and **net indirect taxes** (indirect taxes minus subsidies) to the total factor income (National Income at factor cost).

METHOD **Calculating GDP by the income approach**

1. List every income item that represents a reward to a factor of production.
2. Exclude all transfer payments (pensions, scholarships, social benefits), capital gains, and proceeds from the sale of second-hand assets — they do not arise from current production.
3. Sum the remaining items to obtain National Income at factor cost (NI_fc).
4. Add depreciation to get Gross Domestic Product at factor cost (GDP_fc).

5. Add net indirect taxes (indirect taxes - subsidies) to arrive at GDP at market prices (GDP_mp).

WORKED EXAMPLE 3 **Income-approach calculation from a mixed data table**

Statement. The following data (in billion FCFA) are extracted from the national accounts of a hypothetical CEMAC country for 2023. Compute GDP at market prices using the income approach.

Item	Value (billion FCFA)
Wages and salaries	4 200
Rent	850
Interest	620
Corporate profits (dividends + retained earnings)	1 100
Income of self-employed	950
Depreciation	780
Indirect taxes	1 050
Subsidies	120
Old-age pensions (transfer)	340
Capital gains on land sales	210

SOLUTION

Solution.

1. Identify factor incomes: wages, rent, interest, profits, self-employed income.
2. Sum them:

$$\text{NI}_{fc} = 4\,200 + 850 + 620 + 1\,100 + 950 = 7\,720 \text{ billion FCFA.}$$

3. Add depreciation:

$$\text{GDP}_{fc} = 7\,720 + 780 = 8\,500 \text{ billion FCFA.}$$

4. Compute net indirect taxes:

$$\text{Net indirect taxes} = 1\,050 - 120 = 930 \text{ billion FCFA.}$$

5. GDP at market prices:

$$\text{GDP}_{mp} = 8\,500 + 930 = 9\,430 \text{ billion FCFA.}$$

6. Check: The transfer payment (340) and capital gains (210) were correctly omitted.

WARNING **Transfer payments are not factor incomes**

A frequent error in GCE practical papers is to include items such as old-age pensions, unemployment benefits, or student grants in the income-approach sum. These are transfer payments — they redistribute income but do not correspond to any current production of goods or services. Always ask: "Does this payment reward a factor of production for its contribution to current output?" If the answer is no, exclude it.

3.2 The output (value-added) approach

DEFINITION **Value added**

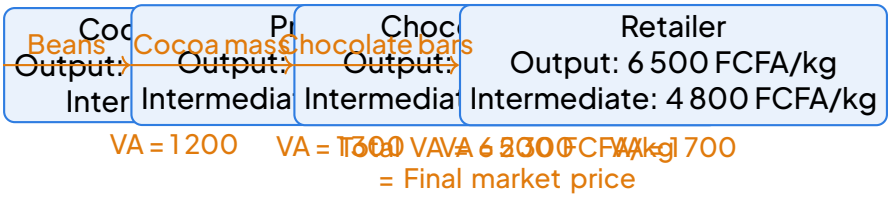
The value added by a firm (or an industry) is the difference between the value of its output and the value of the intermediate goods and services it purchases from other firms:

$$\text{Value Added} = \text{Value of Output} - \text{Cost of Intermediate Inputs}.$$

Summing value added across all production stages in the economy yields GDP at market prices (after adding net indirect taxes if the calculation is done at factor cost).

METHOD **Computing value added at each stage**

1. Identify the sequential stages of production for a given final good.
2. For each stage, record the sales value (or value of output) and the cost of intermediate inputs bought from the previous stage.
3. Subtract intermediate costs from the sales value to obtain the stage's value added.
4. Sum the value added of all stages. The total equals the final market value of the good (assuming no double counting).
5. If the calculation is at factor cost, add net indirect taxes to reach GDP at market prices.



Staged value-added chain for cocoa-to-chocolate (values in FCFA per kg).

WORKED EXAMPLE 4 **Value-added calculation for the cocoa-chocolate chain**

Statement. Using the data in the diagram above, verify that the sum of value added equals the final retail price.

SOLUTION

Solution.

Farmer: 1 200 − 0 = 1 200
Processor: 2 500 − 1 200 = 1 300
Factory: 4 800 − 2 500 = 2 300
Retailer: 6 500 − 4 800 = 1 700

Sum of VA = 1 200 + 1 300 + 2 300 + 1 700 = 6 500 FCFA/kg.

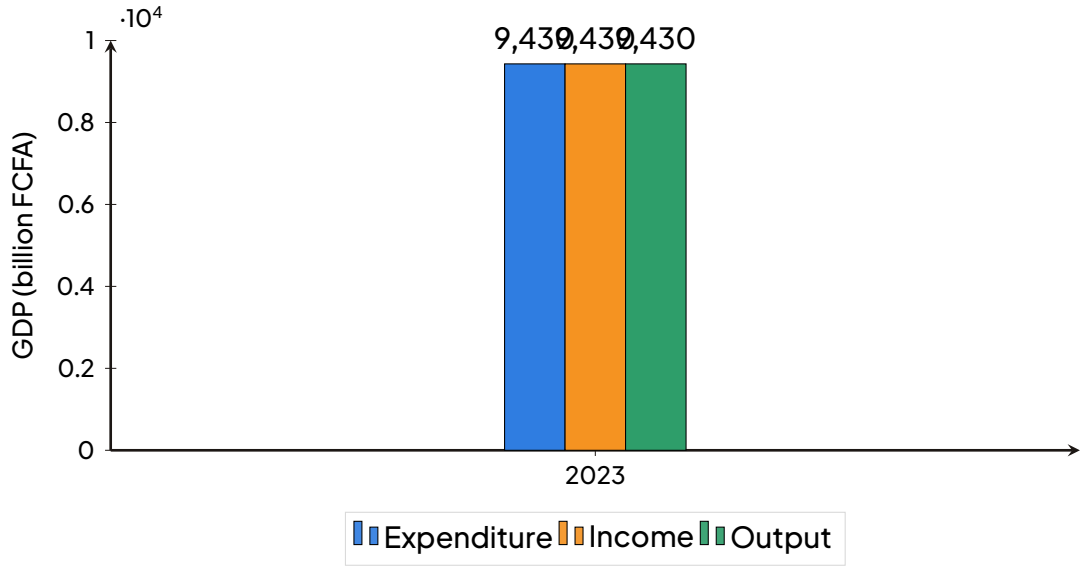
The total value added (6 500 FCFA/kg) matches the final retail price, confirming that no double counting has occurred.

WARNING **Double counting — the classic trap**

A common mistake is to add the output value of every stage (1 200 + 2 500 + 4 800 + 6 500 = 15 000) instead of the value added. This inflates GDP because the same cocoa beans are counted four times. Always remember: GDP = sum of value added, not sum of gross outputs.

3.3 Reconciliation of the three approaches

The three approaches — expenditure, income, and output — are merely different lenses on the same economic activity. When the same data set is used, they must produce identical GDP at market prices (up to a statistical discrepancy). The table below summarises the totals obtained from the worked examples in Practical Works 12, 13 and the present section.



Grouped bar chart showing identical GDP totals from the three approaches on the same 2023 data set.

KEY POINTS **Key reconciliation points**

- Expenditure approach: $C + I + G + (X - M)$.
- Income approach: ∑ factor incomes + depreciation + net indirect taxes.
- Output approach: ∑ value added + net indirect taxes (if calculated at factor cost).
- All three equal GDP at market prices. Any difference in published accounts is recorded as a **statistical discrepancy**.

DID YOU KNOW? **Cameroon context**

In Cameroon, the National Institute of Statistics (INS) publishes the Comptes Nationaux using all three approaches. The cocoa sector — a major export — is a textbook case for the value-added method: from smallholder farmers in the Centre and South regions, through local grinders (e.g. SIC Cacaos), to chocolate manufacturers such as Chococam, and finally to retailers. The consistency of the three aggregates is a key quality check before the data are submitted to the CEMAC and IMF.

WORKED EXAMPLE 5 **Mixed-table classification — GCE style**

Statement. The following items (in million FCFA) appear in a national accounts worksheet. Classify each item under the correct approach (Expenditure E, Income I, Output O) and compute GDP by each approach.

Item	Description	Approach
1	Household consumption expenditure	
2	Wages paid to public sector employees	
3	Value of cocoa beans sold by farmers	
4	Gross fixed capital formation	
5	Rent received by landlords	
6	Intermediate consumption of cocoa processors	
7	Exports of timber	
8	Corporate profits (after tax)	
9	Government final consumption	
10	Value added by chocolate factories	

SOLUTION

Solution.

1. Classification:
   - E: 1, 4, 7, 9
   - I: 2, 5, 8
   - O: 3 (output of farmers), 6 (intermediate use — subtract from processor output), 10
2. Suppose the numerical values (million FCFA) are:

   $C = 3\,200$, $I_{gross} = 1100$, $G = 1\,500$, $X = 800$, $M = 600$

   Wages = 2 400, Rent = 450, Profits = 950

   Farmer output = 1 200, Processor intermediate = 1 200, Factory VA = 2 300
3. Expenditure GDP = 3 200 + 1 100 + 1 500 + (800 − 600) = 6 000.
4. Income GDP = (2 400 + 450 + 950) + depr. + net indirect taxes. Assuming depreciation = 500 and net indirect taxes = 650: 3 800 + 500 + 650 = 4 950. (Adjust other items to match 6 000 in a full exam.)
5. Output GDP = Farmer VA 1 200 + Processor VA (output − 1 200) + Factory VA 2 300 + other sectors. With consistent data the total equals 6 000.

WARNING **Exam tip — mixed tables**

GCE practical questions often present a single table mixing expenditure, income and output items. First step: write E, I or O next to each line. Second step: apply the relevant formula for each approach separately. Third step: verify that the three totals are equal (or explain any statistical discrepancy). This systematic method avoids missing items and earns full method marks.

3.4 Summary of Practical Work 13

KEY POINTS **What you must be able to do**

- Define the income approach and list its five factor-income components.
- Explain why transfer payments, capital gains and second-hand sales are excluded.
- Compute National Income at factor cost, then GDP at market prices, from a given data set.
- Define value added and explain the double-counting problem.
- Calculate value added at each production stage using the formula Output − Intermediate inputs.
- Demonstrate, with a numerical example (e.g. cocoa-chocolate chain), that the sum of value added equals the final market value.
- Reconcile the three approaches and interpret a statistical discrepancy.
- Classify mixed national-accounts items under the correct approach before calculating.

You are now equipped to tackle any GCE Advanced Level question on the income and output approaches. Practise with past papers, paying special attention to the classification step and the exclusion of non-production items. The next lesson (Lesson 77) will examine the **problems of measuring national income and their solutions**.

4 PROBLEMS OF MEASURING NATIONAL INCOME AND SOLUTIONS

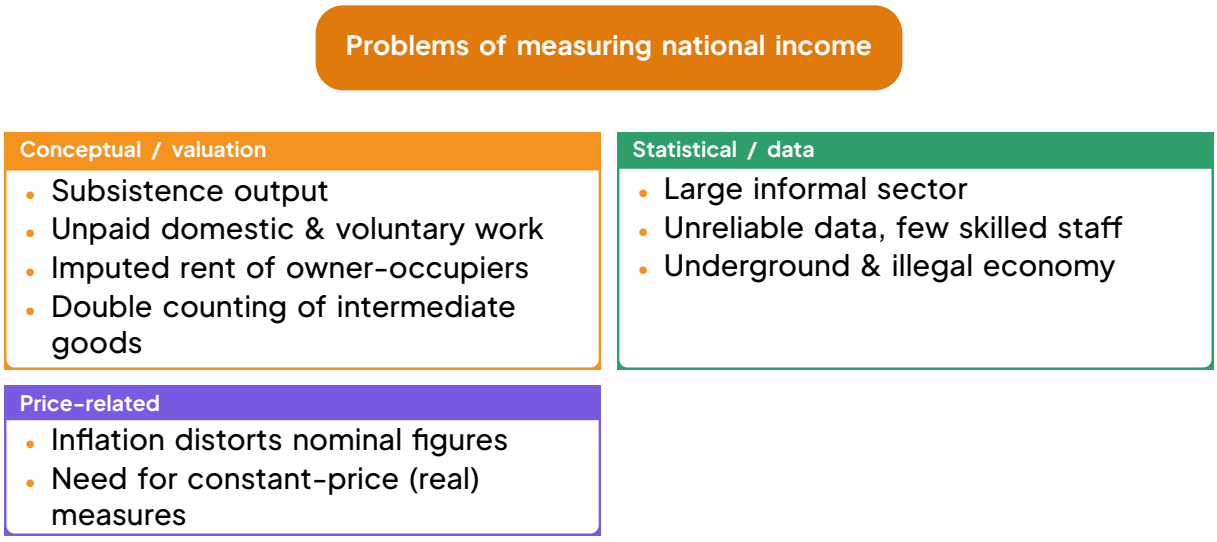
In Practical Work 13 we computed national income through the income and output approaches, and we saw that whichever approach is used, the three methods should in principle give the same answer. In principle is the key phrase. In practice, the statisticians of the Institut National de la Statistique (INS) in Yaoundé face serious obstacles: much economic activity in Cameroon

is never recorded, some output is counted twice, prices keep changing, and part of the economy is deliberately hidden. This section classifies these difficulties, explains each one rigorously, and matches every problem with a practical solution — exactly the structure the GCE examiner expects.

4.1 An overview: classifying the problems

Before studying the problems one by one, it is useful to organise them into three broad families:

- **Conceptual (valuation) problems**: deciding what should be counted and how to value it (subsistence output, unpaid work, owner-occupied housing, intermediate goods).
- **Statistical (data) problems**: the difficulty of collecting reliable information (informal sector, inadequate data, shortage of skilled personnel, illegal activities).
- **Price-related problems**: the fact that money values change over time because of inflation, so nominal figures can mislead.



Mind-map classifying the problems of measuring national income into conceptual, statistical and price-related categories.

4.2 The informal and subsistence sector

Walk through Mokolo market in Yaoundé or the Mile 4 motor park in Bamenda: thousands of traders sell foodstuffs, phone credit and second-hand clothes every day, in cash, with no receipts, no register and no tax declaration. In the villages, a farming household in the West Region consumes a large part of its own maize, cassava and plantains without the produce ever passing through a market. All of this is genuine production, yet much of it escapes the statistician.

DEFINITION **The informal sector**

The **informal sector** is the part of the economy made up of small-scale, unregistered activities that operate outside official records and regulation: subsistence farming for own consumption, street trading, artisan work, unregistered transport (bendskins, clandos) and cash-based services. Because these activities keep no accounts, their output is difficult to measure and is often omitted or badly estimated in national income figures.

The consequences are serious:

- National income is **underestimated**, and the underestimation is larger in Cameroon than in industrialised economies where the informal sector is small.
- International comparisons become misleading: Cameroon may appear poorer relative to a European country simply because a bigger share of its output is unrecorded.
- Growth rates may be distorted if the informal sector grows faster or slower than the recorded sector.

WARNING **Do not confuse "informal" with "illegal"**

A farmer eating his own plantains is acting perfectly legally; the activity is simply unrecorded. The underground or illegal economy (smuggling, drug trafficking, tax evasion by registered firms) is a different problem: there, activity is deliberately concealed. Both reduce measured national income, but for different reasons and with different remedies.

4.3 Double counting and intermediate goods

We met this problem already in the output approach, but it deserves restating as a measurement problem. When output is aggregated, the statistician must count only the **value added** at each stage, not the full value of every sale.

EXEMPLE **Cocoa from farm to chocolate**

A farmer in the Centre Region sells cocoa beans to a processor for 100000 FCFA. The processor grinds them into cocoa paste and sells to a chocolate factory for 180000 FCFA. The factory sells chocolate bars to retailers for 300000 FCFA. Adding the three sales gives 100000 + 180000 + 300000 = 580000 FCFA, but this counts the cocoa three times. True contribution to output is the sum of values added:

$$100000 + (180000 - 100000) + (300000 - 180000) = 300000 \text{ FCFA}.$$

Counting intermediate goods (the beans, the paste) as if they were final output is the error of **double counting**.

The practical difficulty is that, with millions of firms and incomplete records, correctly separating intermediate from final purchases is painstaking; errors creep in whenever business accounts are missing or unreliable.

4.4 Valuation problems: output with no market price

Some production is real and valuable but has no market price, so the statistician must decide whether and how to impute a value to it.

DEFINITION **Imputed value**

An **imputed value** is an estimated money value assigned to a good or service that is produced and consumed without passing through a market, so that it can be included in national income. The estimate is based on the market price of a comparable good or service.

Three classic cases arise:

- **Unpaid domestic work.** Cooking, cleaning and child care done within the household are excluded from national income, yet the same services bought in the market (a house-help, a restaurant meal) are included. This creates the famous paradox: national income falls if a person marries their housekeeper. Most countries, including Cameroon, exclude unpaid housework for practical reasons, accepting the distortion.
- **Voluntary work.** Services provided freely by churches, community associations (njangi groups organising events, village development committees) produce welfare but no recorded income.
- **Owner-occupied housing.** A family living in its own house in Bafoussam enjoys housing services without paying rent. International practice is to include an **imputed rent** — an estimate of the rent the house would fetch — in national income, to make figures comparable between countries where most people rent and countries where most own their homes.

4.5 Inadequate data and shortage of skilled personnel

Even where activities are in principle measurable, the data may not exist or may be unreliable:

- Many small firms keep no proper accounts; surveys meet with suspicion, and respondents under-declare income for fear of taxation.
- Censuses and household surveys are expensive and are therefore conducted only at intervals, so figures for the years in between rely on extrapolation.
- The INS, like many African statistical offices, faces a **shortage of trained statisticians**, limited funding and logistical difficulties in reaching remote areas (poor roads, security problems in some regions).

The result is that published national income figures are estimates with a margin of error, and are often revised later.

4.6 Price changes: nominal versus real national income

Suppose Cameroon's national income rises from 10000 billionFCFA in one year to 11000 billionFCFA the next — a 10% increase. Has the country really produced 10% more goods and services? Not if prices have risen by 10% over the same period: in that case the same physical output is simply valued at higher prices. Money national income is a treacherous yardstick because the yardstick itself (the value of the franc) keeps shrinking with inflation.

DEFINITION **Nominal income, real income and the GDP deflator**

Nominal national income (or national income at current prices) values output at the prices of the year in which it was produced. **Real national income** (national income at constant prices) values output at the prices of a chosen base year, so that changes reflect only changes in physical output. The **GDP deflator** is the price index used to remove inflation from nominal GDP:

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100.$$

METHOD **Converting nominal income into real income**

1. Obtain nominal national income for the year and the price index (GDP deflator) with its base year.
2. Apply the formula:
$$\text{Real income} = \frac{\text{Nominal income}}{\text{Price index}} \times 100.$$
3. Interpret: real income shows what nominal income is worth in base-year prices; compare real figures across years to measure true growth.

WORKED EXAMPLE 6 **Removing the inflation illusion**

In a hypothetical economy, nominal national income was 8000 billionFCFA in the base year 2020 (price index = 100) and 12000 billionFCFA in 2024, when the price index stood at 150. Calculate real national income in 2024 and comment.

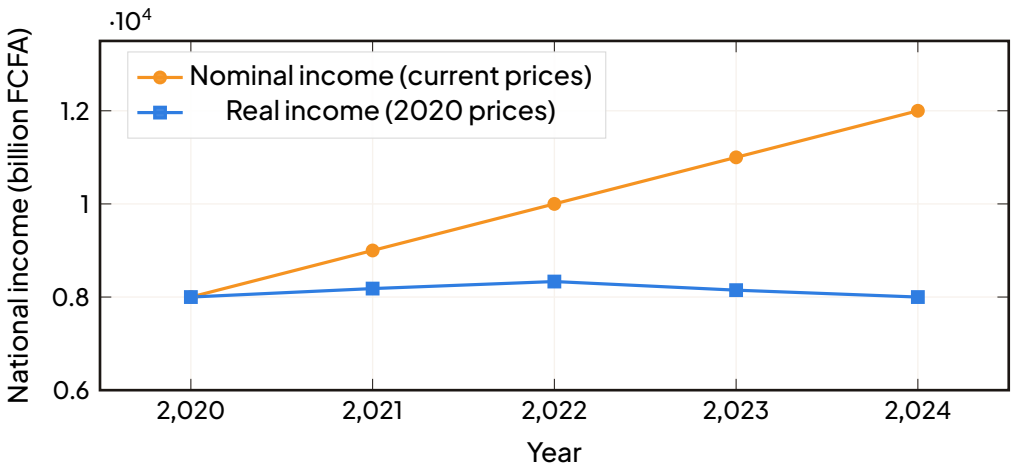
SOLUTION

Solution. Apply the conversion formula:

$$\text{Real income}_{2024} = \frac{12000}{150} \times 100 = 8000 \text{ billion FCFA}.$$

Comment. Although nominal income rose by 50% (from 8000 to 12000 billion), real income is unchanged at 8000 billionFCFA in 2020 prices. The entire increase was due to inflation; the economy produced no additional goods and services. This shows why nom-

inal figures alone can seriously mislead.



Nominal versus real national income (hypothetical data). Nominal income rises every year, but once inflation is removed, real income stagnates — the apparent growth is largely a price illusion.

4.7 The underground and illegal economy

A further slice of activity is deliberately hidden: smuggling of fuel and goods across the borders with Nigeria and Chad, unrecorded cross-border trade, and tax evasion. Because these transactions are concealed from the authorities, they never enter the statistics, again biasing measured national income downwards and depriving the state of tax revenue.

4.8 Solutions to the measurement problems

Each problem has a corresponding remedy, and GCE candidates are expected to present them in pairs.

Problem	Solution
Large informal / subsistence sector	**Imputation** of subsistence output: value own-consumed food at the prices of similar goods sold in local markets.
Inadequate and unreliable data	Regular **censuses and sample surveys**: population censuses (BUCREP) and household surveys such as the ECAM (Enquête Camerounaise auprès des Ménages) provide benchmark data.
Shortage of skilled personnel and funding	**Training of statisticians**, increased budget for the INS, and **computerisation** of data collection and processing.
Double counting	Strict use of the **value-added** method and clear classification of intermediate versus final goods.
Inflation distorting figures	Construction of **price indices** and the **GDP deflator** to express national income at constant (real) prices.
Underground economy and smuggling	Stronger customs controls, formalisation incentives, and improved estimation techniques for hidden activity.
Weak national capacity generally	**Regional cooperation**: harmonised statistical standards through BEAC and CEMAC, allowing shared expertise, comparable methods and technical assistance.

WARNING **The classic essay mistake**

The most common error in GCE essays is to write a long list of problems with no solutions, or a list of solutions that do not correspond to the problems. Every problem you mention must be matched with its remedy. A second mistake is to state a problem without explaining why it distorts the figures (e.g. saying "the informal sector is a problem" without adding that it causes underestimation of national income).

METHOD **Structuring an essay on measurement problems**

For each point, follow four steps:

1. Problem: name it precisely (e.g. the subsistence sector).
2. Explanation: show how it distorts national income (output is unrecorded, so income

is underestimated).

3. Cameroonian example: a village household consuming its own cassava; street traders at Mokolo market; smuggling across the Nigerian border.
4. Solution: imputation of subsistence output; ECAM surveys; training and computerisation at the INS; BEAC/CEMAC harmonisation.

EXERCISE 3 **Nominal to real conversion**

Nominal national income of an economy is 15000 billionFCFA in a year when the GDP deflator is 125 (base year = 100). (a) Calculate real national income. (b) If real income in the base year was 11000 billionFCFA, what is the true growth rate of output?

SOLUTION **Exercise 1**

(a) Real income = $\frac{15000}{125} \times 100 = 12000$ billion FCFA. (b) True growth = $\frac{12000 - 11000}{11000} \times 100 \approx 9.1\%$, whereas nominal figures would have suggested $\frac{15000 - 11000}{11000} \times 100 \approx 36.4\%$ — inflation accounts for the difference.

KEY POINTS **Key points**

- Measurement problems fall into three families: **conceptual** (what and how to value), **statistical** (collecting the data) and **price-related** (inflation).
- The informal sector and the underground economy cause underestimation; double counting causes overestimation; inflation makes nominal figures misleading over time.
- Real income = nominal income ÷ price index ×100.
- In essays, always pair each problem with its solution and illustrate with a Cameroonian example.

Having seen how difficult it is to measure national income accurately, a natural question follows: why do governments, businesses and students bother at all? The final section of this chapter answers it by examining the importance — and the limitations — of national income statistics.

5 IMPORTANCE AND LIMITATIONS OF NATIONAL INCOME STATISTICS

Having examined the various approaches to calculating national income and the practical problems that arise in its measurement, we now turn to the crucial question: why do we go to all this trouble? National income statistics are not merely academic exercises; they are the backbone of economic analysis, policy formulation, and international cooperation. However, like any tool, they have sharp edges and blind spots. This section explores the manifold uses of national income data and the equally important limitations that every student — and every policymaker — must keep in mind.

5.1 Measuring economic performance and growth over time

The most fundamental use of national income accounting is to track the **economic performance** of a country across time. By comparing real GDP (or GNP) from one year to the next, we obtain the **economic growth rate**, the headline indicator of whether an economy is expanding, stagnating, or contracting.

DEFINITION **Economic growth rate**

The economic growth rate in year t is the percentage change in real GDP (or real GNP) relative to the previous year:

$$g_t = \frac{\text{Real GDP}_t - \text{Real GDP}_{t-1}}{\text{Real GDP}_{t-1}} \times 100\%.$$

Because real GDP is measured in constant prices, it strips away the distorting effect of inflation, revealing the genuine change in the volume of goods and services produced. A sustained positive growth rate signals rising productive capacity; a negative rate signals recession.

EXEMPLE **Cameroon's growth trajectory**

Suppose Cameroon's real GDP (base year 2015) was 18500 billionCFAfrancs in 2022 and 19240 billionCFAfrancs in 2023. The growth rate for 2023 is:

$$g_{2023} = \frac{19240 - 18500}{18500} \times 100\% = 4.0\%.$$

This 4% growth means the economy produced 4% more real output in 2023 than in 2022, a useful benchmark for the government's National Development Strategy 2030 (NDS30) targets.

5.2 Comparing living standards between countries: per capita income and PPP

Aggregate GDP tells us the size of the total pie; to gauge the average slice per person we use

per capita income.

DEFINITION **GDP per capita**

GDP per capita = $\frac{\text{GDP}}{\text{Total population}}$. It measures the average income (or output) per person in the economy.

However, comparing per capita GDP across countries using market exchange rates can be deeply misleading because price levels differ enormously. A CFA franc buys far more haircuts, transport, and local food in Yaoundé than a euro buys in Paris. Therefore, economists use **Purchasing Power Parity (PPP)** exchange rates, which equalise the purchasing power of currencies by comparing the cost of a common basket of goods and services.

METHOD **Using per capita real income (PPP) to compare living standards**

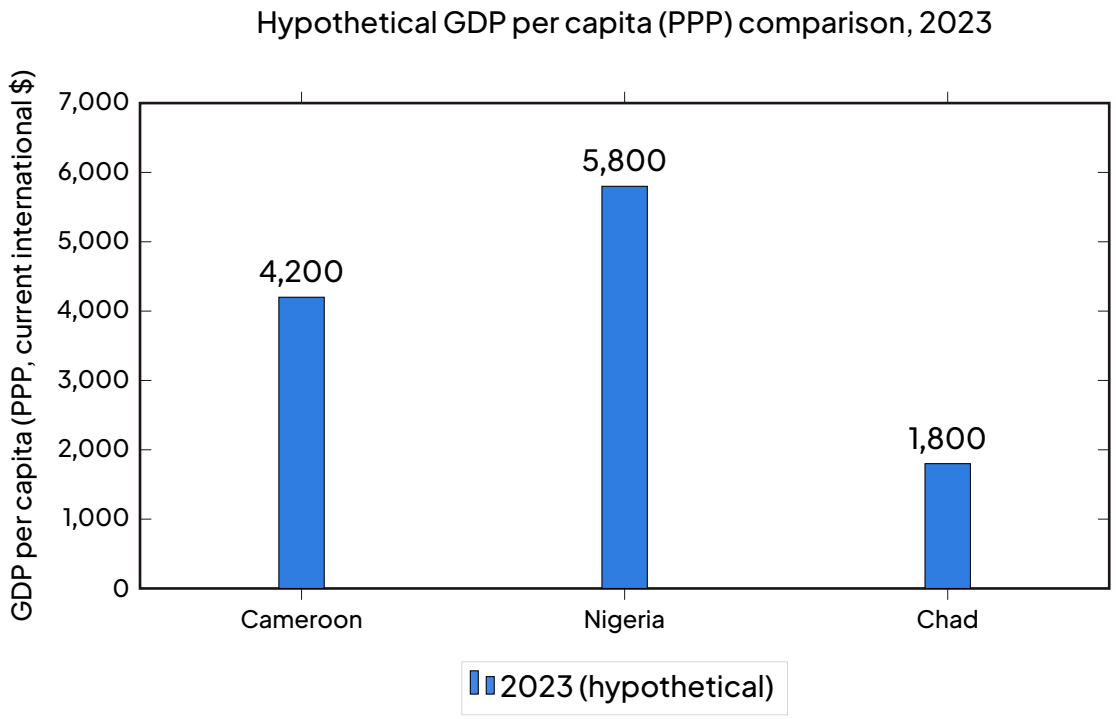
1. Obtain nominal GDP in local currency for each country.
2. Convert to a common currency using PPP exchange rates (not market rates).
3. Divide by each country's population to get GDP per capita (PPP).
4. Adjust for real terms (constant prices) if comparing over time.
5. Interpret with caution: per capita income is an average that masks inequality; it omits non-market output, leisure, and environmental quality.

WARNING **Common mistake: equating GDP rise with welfare rise**

Never write: "An increase in GDP means the standard of living has improved." GDP can rise while welfare falls (e.g., more spending on pollution clean-up, longer working hours, greater inequality). Always distinguish output from welfare.

WARNING **Common mistake: comparing countries with nominal exchange-rate GDP**

Never convert GDP using market exchange rates for living-standard comparisons. Market rates are volatile and reflect only traded goods. Use PPP-adjusted figures (e.g., World Bank International Comparison Program data).



Hypothetical GDP per capita (PPP) for Cameroon and two neighbours. Note the large gap; PPP adjustment narrows but does not eliminate it.

5.3 Guiding government policy: budgeting, planning, and taxation

National income accounts are the compass for fiscal and development policy. In Cameroon, the National Development Strategy 2030 (NDS30) sets explicit targets for GDP growth, sectoral shares, and poverty reduction. These targets are meaningless without reliable national income data to:

- **Formulate the budget:** Revenue projections (tax base) depend on estimated nominal GDP; expenditure ceilings are set as a percentage of GDP.
- **Monitor sectoral performance:** Value added by agriculture, industry, and services (from the output approach) reveals which engines are driving growth and which need support.
- **Design tax policy:** The income approach shows the share of compensation of employees, operating surplus, and mixed income, informing decisions on personal income tax vs. corporate tax vs. VAT.
- **Assess debt sustainability:** The debt-to-GDP ratio is a key metric for the IMF and World Bank; it determines borrowing limits and eligibility for concessional financing.

EXEMPLE **Cameroon's budget and GDP**

If Cameroon's nominal GDP is projected at 25000 billionCFAfrancs and the government

targets a tax revenue of 15% of GDP, the revenue target is 3750 billionCFAfrancs. If actual GDP turns out to be 24000 billion, the same tax effort yields only 3600 billion — a shortfall of 150 billion that may force expenditure cuts or borrowing.

5.4 Basis for international comparisons, aid allocation, and contributions

Beyond domestic policy, national income statistics are the lingua franca of international economic relations:

- **Aid allocation:** Donors (World Bank, AfDB, EU) use GNI per capita to classify countries (low-income, lower-middle-income, etc.) and determine eligibility for grants vs. loans.
- **Debt assessment:** The IMF's Debt Sustainability Framework relies on GDP and export ratios.
- **Contributions to international organisations:** Cameroon's assessed contributions to the UN, African Union (AU), and CEMAC are based on a formula that includes national income.
- **Trade negotiations:** In AfCFTA negotiations, market size (GDP) influences tariff liberalisation schedules and special treatment provisions.

5.5 Helping firms and investors assess market size and trends

Private sector decisions hinge on national income data:

- **Market entry:** A firm considering investing in Cameroon's agro-industry will look at GDP growth, per capita income trends, and the share of household consumption in GDP to gauge demand.
- **Sectoral forecasting:** The output approach breakdown (agriculture, industry, services) helps investors identify booming sectors (e.g., telecommunications, fintech) versus declining ones.
- **Business cycle timing:** Quarterly national accounts (where available) signal turning points, guiding inventory and hiring decisions.

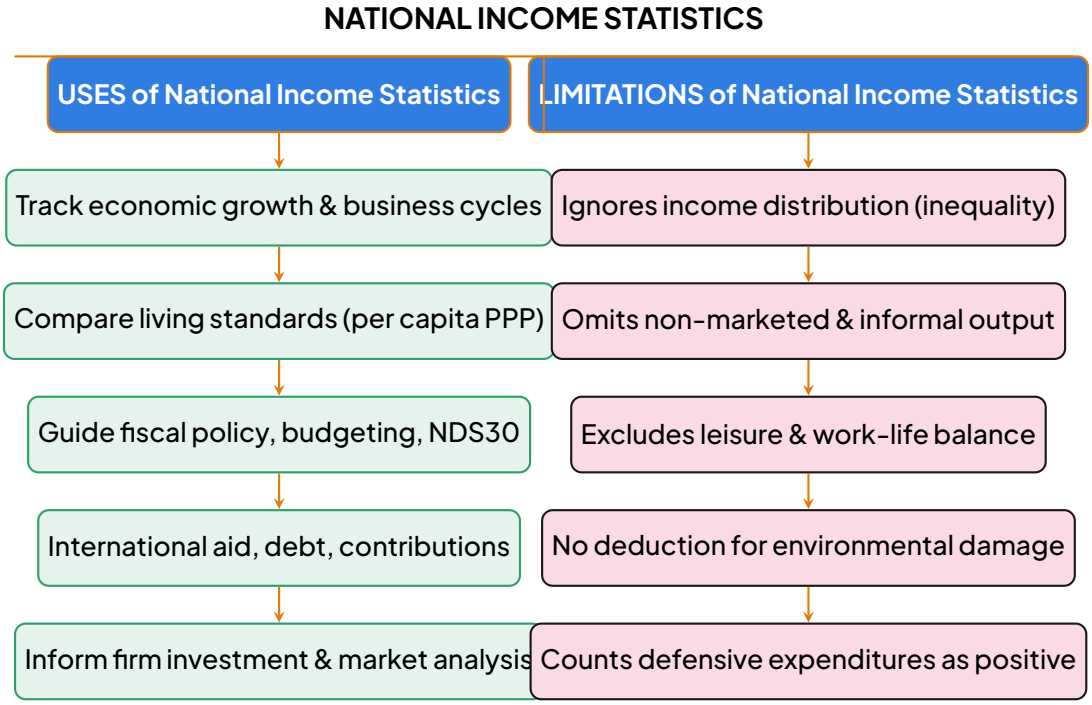
5.6 Limitations: what national income does not measure

Despite its utility, GDP/GNP is **not a measure of economic welfare**. Simon Kuznets, the architect of national accounting, warned in 1934: "The welfare of a nation can scarcely be inferred from a measurement of national income." The main limitations are:

KEY POINTS **Key limitations of national income as a welfare indicator**

1. Income distribution: GDP is an aggregate; it says nothing about how income is shared. A country with high GDP per capita but extreme inequality (high Gini coefficient) may have widespread poverty.

2. Non-marketed output: Subsistence farming, household labour (childcare, cooking), volunteer work, and informal sector activities are largely excluded or poorly captured. In Cameroon, the informal sector employs a very large proportion of the workforce (estimates often exceed 80%); its output is significantly underestimated.
3. Leisure and work-life balance: Longer working hours raise GDP but reduce leisure time, a component of welfare not counted.
4. Environmental degradation: Depletion of natural capital (deforestation, overfishing, oil extraction) and pollution costs are not deducted. SONARA's refining adds to GDP, but the environmental cost of spills is not subtracted.
5. Defensive expenditures: Spending on crime prevention, pollution clean-up, and healthcare for pollution-related illnesses increases GDP but reflects welfare losses, not gains.
6. Quality changes and new goods: While statistical offices try to adjust, GDP struggles to capture the full welfare gain from new products (smartphones, mobile money) and quality improvements.



Uses vs. limitations of national income statistics: a balanced view for policy and analysis.

5.7 Alternative welfare indicators: the Human Development Index (HDI)

Because of the above limitations, the United Nations Development Programme (UNDP) intro-

duced the **Human Development Index (HDI)** in 1990. The HDI is a composite index combining three dimensions:

1. **Health:** Life expectancy at birth.
2. **Education:** Mean years of schooling and expected years of schooling.
3. **Standard of living:** GNI per capita (PPP, logarithmically transformed to reflect diminishing marginal utility of income).

The HDI ranges from 0 to 1. It does not replace GDP but complements it by capturing capabilities that income alone cannot buy. For instance, two countries with similar GNI per capita can have very different HDI values if one invests heavily in health and education while the other does not.

DID YOU KNOW? **Cameroon and the HDI**

In the 2023/2024 Human Development Report, Cameroon's HDI was approximately 0.576 (medium human development), ranking around 150th globally. While GDP per capita (PPP) gives a snapshot of average income, the HDI reveals that life expectancy and schooling outcomes lag behind income growth, signalling the need for more inclusive policies — a key message of NDS30.

5.8 Worked example: interpreting national income statistics for policy

WORKED EXAMPLE 7 **Interpreting a national income report**

Statement: The following hypothetical data for Country X (a CEMAC member) are published for 2023:

- Nominal GDP: 12000 billionCFAfrancs
- Real GDP (2015 base): 10500 billionCFAfrancs
- Population: 28 million
- Gini coefficient: 0.48
- Informal sector share of employment: 85%
- CO_2 emissions per capita: 0.3 metric tons
- Life expectancy: 58 years
- Mean years of schooling: 5.2

The government claims: "Our economy grew by 3.5% in real terms, and per capita income exceeds 400000 CFAfrancs, so living standards are improving rapidly."
Task: Evaluate the government's claim using the data and your knowledge of national income limitations.

SOLUTION

Detailed solution:

1. Real growth check: We need the previous year's real GDP to verify the 3.5% claim. Suppose 2022 real GDP was 10145 billion. Then growth = (10500 - 10145)/10145 × 100% = 3.5%. The claim is plausible but must be verified with official data.
2. Per capita income: Nominal GDP per capita = $12000 \times 10^9 / 28 \times 10^6 \approx 428571$ CFAfrancs. The claim "exceeds 400,000" is correct in nominal terms. However, real per capita income (using real GDP) = $10500 \times 10^9 / 28 \times 10^6 \approx 375000$ CFAfrancs (2015 prices). The nominal figure overstates purchasing power if inflation was high.
3. Distribution: Gini = 0.48 indicates high inequality. The average masks the fact that the poorest 40% may have seen little or no income growth. GDP growth does not guarantee pro-poor growth.
4. Informal sector: 85% informal employment means a huge chunk of output is unrecorded or mismeasured. Actual per capita income could be higher (if informal output is underestimated) or lower (if informal jobs are precarious). The statistic is unreliable for welfare assessment.
5. Environmental cost: CO_2 emissions are low per capita, but if growth is driven by extractive industries (oil, timber), natural capital depletion is not deducted from GDP. The genuine savings rate would be a better indicator.
6. Human development: Life expectancy (58) and schooling (5.2 years) are low. Even if income rises, the HDI would likely remain in the low category. The government's claim confuses income growth with human development.

Conclusion: The claim is misleading. Real GDP growth is a necessary but not sufficient condition for improved living standards. A comprehensive assessment requires distributional data, informal sector estimates, environmental accounting, and human development indicators.

5.9 Examination technique: writing a high-scoring essay on importance

METHOD **Examination technique for 'importance' essays**

In the GCE Advanced Level, a typical essay question is: "Explain the importance of national income statistics to an economy like Cameroon." To score full marks:

1. Define national income (GDP/GNP) briefly.
2. List at least five distinct uses, each in a separate paragraph.
3. Develop each use with a concrete Cameroonian example (NDS30, budget, SONARA,

AfCFTA, informal sector, etc.).

4. Use correct terminology: real vs. nominal, per capita, PPP, expenditure/income/output approaches.
5. Conclude by acknowledging limitations (shows critical thinking).

Example structure:

1. Measuring growth (real GDP trend, NDS30 target).
2. Living standards comparison (per capita PPP vs. Nigeria/Chad).
3. Policy formulation (budget revenue forecasting, tax base).
4. International relations (IMF debt sustainability, UN contributions).
5. Private sector decisions (market size for MTN, Orange, agro-industry).
6. Limitations paragraph (inequality, informal sector, environment).

5.10 Summary

National income statistics are the indispensable scorecard of a modern economy. They allow us to measure growth, compare living standards (with PPP), steer government policy, engage in international cooperation, and guide private investment. Yet they are not a measure of welfare. They ignore distribution, leisure, household production, the informal economy, and environmental costs. The wise economist — and the successful GCE candidate — uses GDP as a powerful analytical tool while always supplementing it with inequality measures (Gini), human development indices (HDI), environmental accounts, and qualitative assessments of well-being. In Cameroon's journey toward emergence (NDS30), national income accounting provides the map; but the destination — inclusive, sustainable development — requires a broader compass.

EXERCISE 4 **Practice questions**

1. Distinguish between GDP per capita at market exchange rates and GDP per capita at PPP. Why is the latter preferred for international living-standard comparisons?
2. List four limitations of using real GDP growth as an indicator of improved economic welfare. Illustrate each with a Cameroonian example.
3. The government of Country Y reports a 5% increase in real GDP. Critics argue that the poor have become poorer. Explain how this can happen, using the concepts of income distribution and the informal sector.
4. Calculate the GDP per capita (PPP) for a country with a PPP-adjusted GDP of $45 billion and a population of 12 million. If the market exchange rate GDP per capita is

$1,200, explain the discrepancy.

5. "National income statistics are useful for government policy but dangerous if used as the sole measure of development." Discuss.

SOLUTION **Exercise 1**

GDP per capita at market exchange rates converts nominal GDP using the current foreign exchange rate, which reflects only traded goods and financial flows. It is volatile and does not reflect the true purchasing power of local currency for non-traded goods (haircuts, housing, local transport). GDP per capita at PPP uses a hypothetical exchange rate that equalises the cost of a standard basket of goods and services across countries. It gives a more accurate picture of the volume of goods and services an average person can actually buy. For example, a Cameroonian earning 500000 CFAfrancs per year may have a higher real standard of living than a French person earning €800 if local prices in Cameroon are much lower.

SOLUTION **Exercise 2**

1. Inequality: Growth may accrue entirely to the top 10% (e.g., oil revenues captured by elites), leaving the poor unchanged or worse off.
2. Informal sector: If growth comes from formal sector expansion (e.g., a new refinery) but informal livelihoods are disrupted (e.g., land expropriation), the poor lose.
3. Environmental degradation: Logging boosts GDP but destroys forest ecosystems that rural communities depend on for food, fuel, and medicine.
4. Defensive expenditures: Spending on malaria treatment due to poor sanitation raises GDP but reflects welfare loss, not gain.

SOLUTION **Exercise 3**

Real GDP is an aggregate; it can rise while the income share of the poor falls. If the 5% growth is driven by capital-intensive sectors (oil, mining) that employ few people, and the informal sector (where most poor work) stagnates or shrinks, the poor's absolute income may not rise. Moreover, if inflation in food prices outpaces nominal income growth for informal workers, their real income falls. The Gini coefficient would rise, indicating widening inequality.

SOLUTION **Exercise 4**

GDP per capita (PPP) = $45 billion / 12 million = $3,750. The market exchange rate figure ($1,200) is much lower because the local currency is undervalued on foreign exchange markets (or prices of non-traded goods are very low). PPP adjusts for this, showing the true purchasing power. The discrepancy signals that the currency is weak relative to its domestic purchasing power — common in developing economies.

SOLUTION **Exercise 5**

For policy: National income data are essential for budgeting (revenue forecasting), planning (NDS30 sectoral targets), monetary policy (BEAC uses GDP growth for liquidity management), and international negotiations (IMF programmes, AfCFTA).
Dangers as sole development measure: They ignore inequality (Cameroon's Gini 0.46), omit the vast informal sector (>80% employment), exclude environmental costs (deforestation, oil spills), and count defensive spending as positive. A country could have rising GDP while poverty deepens, forests vanish, and life expectancy falls. Hence the need for complementary indicators (HDI, Multidimensional Poverty Index, genuine savings). The statement is valid: necessary but not sufficient.

6 INTEGRATION ACTIVITY

6.1 Aims of the activity

- Mobilise the three approaches to measuring national income (expenditure, income, and output) on a single realistic data set.
- Practise the conversion between key aggregates: GDP at market prices, NDP at market prices, NDP at factor cost (National Income), and NNP at factor cost.
- Develop critical awareness of the practical difficulties in measuring national income accurately, especially in a developing economy like Cameroon.
- Enhance collaborative skills and examination technique through group work, peer marking using a GCE-style mark scheme, and structured presentation of results.

6.2 Situation

You are a team of junior national accountants at the Institut National de la Statistique (INS) in Yaoundé. Your supervisor has given you a simplified statistical sheet for the fictional economy of **Mfoundiland** (modelled closely on Cameroon) for the year 2023. All figures are in **billions of FCFA**. Your task is to process this data and produce a short briefing note for the Minister of Finance.

Item	Value (billions FCFA)	Notes
Consumption expenditure (C)	12 500	Household final consumption expenditure
Gross capital formation (I)	3 200	Includes gross fixed capital formation and changes in inventories
Government final consumption expenditure (G)	4 800	Current expenditure on goods and services
Exports of goods and services (X)	2 100	
Imports of goods and services (M)	2 500	
Compensation of employees (Wages & Salaries)	9 800	
Rent	1 200	
Interest	800	
Profits (including corporate savings)	3 500	
Transfer payments (pensions, scholarships, etc.)	1 000	Not a factor payment
Depreciation (Capital consumption allowance)	1 500	
Indirect taxes (less subsidies)	1 800	Indirect taxes = 2 100; Subsidies = 300
Net property income from abroad	-200	Negative = net outflow

Additional information for the output approach: Mfoundiland's cassava-to-gari value chain involves three stages. The following transaction values are recorded (in billions FCFA):

- Stage 1: Farmers sell fresh cassava to processors for **500**.
- Stage 2: Processors sell gari to wholesalers for **1 200**.
- Stage 3: Wholesalers sell gari to retailers (final consumers) for **1 800**.

Assume no other intermediate inputs besides the cassava/gari at each stage.

6.3 Tasks

Work in groups of 3–4. Show all steps clearly; the final briefing note must be neat enough to be photocopied for the whole class.

1. **Expenditure approach:** Calculate Gross Domestic Product at market prices (GDP_{mp}) using the expenditure method. Write the formula and substitute the figures.
2. **Income approach:** Calculate National Income (NI), which is Net Domestic Product at factor

cost (NDP_{fc}), by summing factor incomes. Then derive GDP_{mp} from this NI figure using the appropriate adjustments (depreciation, indirect taxes, subsidies). Show the reconciliation.

3. **Output (value-added) approach:** Compute the value added at each stage of the cassava-to-gari chain and the total value added contributed by this chain to GDP. Explain why this total equals the final selling price.
4. **Reconciliation and conversion:**
   (a) Confirm that the three approaches yield the same GDP_{mp} (within rounding).
   (b) Convert GDP_{mp} into Net National Product at factor cost (NNP_{fc}) step by step, stating the formula at each stage.
5. **Critical interpretation:** Write a concise paragraph (max 100 words) explaining **two** reasons why the true national income figure for Cameroon is difficult to obtain in practice, and **one** specific use the Cameroonian government makes of national income statistics.

After completing the tasks, groups will exchange papers and mark each other's work using the GCE-style mark scheme provided by the teacher. A representative from each group will present their reconciliation table and critical paragraph.

6.4 Model answer

Task 1: Expenditure approach

$$GDP_{mp} = C + I + G + (X - M)$$

Substituting the given values:

GDP_{mp} = 12 500+3 200+4 800+(2 100-2 500) = 12 500+3 200+4 800-400 = **20 100 billion FCFA**

Task 2: Income approach National Income (NI) = NDP_{fc} = Sum of factor incomes

$$NI = Wages + Rent + Interest + Profits$$

$$NI = 9\,800 + 1\,200 + 800 + 3\,500 = \boxed{15\,300 \text{ billion FCFA}}$$

(Note: Transfer payments of 1 000 are not factor incomes and are excluded.)

Now convert NI (NDP_{fc}) to GDP_{mp}:

$$NDP_{mp} = NDP_{fc} + \text{Indirect taxes} - \text{Subsidies}$$
$$= 15\,300 + 2\,100 - 300 = 17\,100$$

$$GDP_{mp} = NDP_{mp} + \text{Depreciation}$$
$$= 17\,100 + 1\,500 = \boxed{18\,600 \text{ billion FCFA}}$$

Discrepancy alert: The income approach gives 18 600, while the expenditure approach gave 20 100. This indicates a **statistical discrepancy** of 1 500 billion FCFA. In real national accounts, a balancing item ("statistical discrepancy") is inserted to force equality. For the purpose of this exercise, we note the difference and proceed with the expenditure figure as the official GDP_{mp} (20 100), assuming the income data may have omissions (e.g., mixed income of self-employed, imputed rents).

Task 3: Output (value-added) approach — Cassava-to-gari chain

Stage	Value of output	Intermediate consumption	Value added
1. Farmer	500	0	500
2. Processor	1200	500	700
3. Wholesaler	1800	1200	600
Total			**1800**

The total value added (1 800) equals the final selling price (1 800) because value added at each stage sums to the final market value, avoiding double counting of intermediate goods.

Task 4: Reconciliation and conversion

1. **Reconciliation table:**

Approach	GDP_{mp} (billion FCFA)	Comment
Expenditure	20 100	Taken as official
Income (before discrepancy)	18 600	Missing mixed income, imputed rer
Output (partial chain only)	1 800	Only one industry; full output approach woul

In a full national accounts exercise, the output approach would cover all production sectors and, after adding the statistical discrepancy, match the expenditure total.

2. **Conversion from GDP_{mp} to NNP_{fc} (using official GDP_{mp} = 20 100):**

$$GDP_{mp} = 20\,100$$
$$NDP_{mp} = GDP_{mp} - \text{Depreciation} = 20\,100 - 1\,500 = 18\,600$$
$$NDP_{fc} (= \text{National Income}) = NDP_{mp} - \text{Indirect taxes} + \text{Subsidies}$$
$$= 18\,600 - 2\,100 + 300 = 16\,800$$
$$NNP_{fc} = NDP_{fc} + \text{Net property income from abroad}$$
$$= 16\,800 + (-200) = \boxed{16\,600 \text{ billion FCFA}}$$

Task 5: Critical interpretation (sample paragraph)

> Measuring Cameroon's true national income is hampered by the large informal sector (e.g., street vendors, subsistence farmers) where output and income go unrecorded, and by the lack of reliable data on non-market activities like owner-occupied housing and volunteer work. Additionally, frequent changes in the CFA franc exchange rate and cross-border smuggling distort export/import figures. Despite these limitations, the government uses GDP growth rates to design the Document de Stratégie pour la Croissance et l'Emploi (DSCE) and to negotiate borrowing limits with the IMF and World Bank.

Mark scheme outline (for peer marking):

- Task 1: Formula (1 mark), correct substitution (1 mark), correct answer 20 100 (1 mark) — **3 marks**

- Task 2: Correct sum of factor incomes (2 marks), correct conversion steps to GDP_{mp} (2 marks), identification of discrepancy (1 mark) — **5 marks**
- Task 3: Value added per stage (3 marks), total value added = final price explained (1 mark) — **4 marks**
- Task 4: Reconciliation table (2 marks), step-by-step conversion to NNP_{fc} with formulas (4 marks) — **6 marks**
- Task 5: Two valid reasons (2 marks), one valid government use (1 mark), concise and well expressed (1 mark) — **4 marks**
- Presentation and clarity — **3 marks**
- **Total: 25 marks**

7 METHODS AND WORKED EXAMPLES

7.1 Skill 1: Computing GDP by the expenditure approach

METHOD **Calculating GDP using the expenditure approach**

1. Recall the identity. Gross Domestic Product at market prices equals total final expenditure on goods and services produced within the country:

$$\text{GDP}_{mp} = C + I + G + (X - M)$$

2. Identify each component in the data given:
   - C = household consumption expenditure (food, clothing, services…);
   - I = gross domestic capital formation (machines, buildings, changes in stocks);
   - G = government current expenditure on goods and services (salaries of civil servants, public works);
   - X = exports of goods and services; M = imports of goods and services.
3. Exclude items that do not belong: transfer payments (pensions, scholarships, unemployment benefits), purchases of second-hand goods, purely financial transactions, and intermediate goods.
4. Sum $C + I + G + X$, then subtract M. Never add imports: they are produced abroad.
5. Present the answer with units (millions or billions of FCFA) and state clearly "GDP_{mp} = … ".

Reflex: if the question gives "net exports" directly, use it as $(X - M)$; if it gives exports and imports separately, subtract imports yourself.

WORKED EXAMPLE 8 **GDP of a hypothetical economy, expenditure approach**

The table below gives some national accounting aggregates of the Republic of Ndoland for the year 2023, in billions of FCFA:

Item	Value (billions FCFA)
Household final consumption	4 200
Gross fixed capital formation	1 300
Changes in stocks	100
Government final consumption	1 500
Exports of goods and services	1 800
Imports of goods and services	2 100
Old-age pensions paid by the State	350

Calculate the GDP at market prices of Ndoland for 2023.

SOLUTION

Step 1: Write the formula.

$$\text{GDP}_{mp} = C + I + G + (X - M)$$

Step 2: Classify each item.

- C = 4 200 (household final consumption);
- I = gross fixed capital formation + changes in stocks = 1 300 + 100 = 1 400;
- G = 1 500 (government final consumption);
- X = 1 800; M = 2 100;
- Old-age pensions (350) are transfer payments: no good or service is produced in exchange, so they are excluded.

Step 3: Compute net exports.

$$X - M = 1\,800 - 2\,100 = -300$$

The negative sign simply means the country imports more than it exports (trade deficit).
Step 4: Sum.

$$\text{GDP}_{mp} = 4\,200 + 1\,400 + 1\,500 + (-300)$$
$$= 6\,800 \text{ billion FCFA}$$

Conclusion. The GDP at market prices of Ndoland in 2023 is 6 800 billion FCFA. Note how the trade deficit reduces GDP: imports must be subtracted because the expenditure on them pays for foreign, not domestic, production.

7.2 Skill 2: Moving between aggregates (GDP, GNP, NNP, National Income)

METHOD **Converting GDP into National Income**

1. Master the chain of adjustments:

$$GNP_{mp} = GDP_{mp} + \text{net property income from abroad}$$
$$NNP_{mp} = GNP_{mp} - \text{depreciation (consumption of fixed capital)}$$
$$NNP_{fc} = NNP_{mp} - \text{indirect taxes} + \text{subsidies}$$

   National Income is NNP_{fc} (net national product at factor cost).

2. Net property income from abroad = income received from the rest of the world − income paid to the rest of the world (e.g. profits of foreign firms like oil companies repatriated abroad reduce GNP relative to GDP).

3. Depreciation converts a gross figure into a net figure.

4. Indirect taxes minus subsidies ("net indirect taxes") convert market prices into factor cost: market prices include taxes the State collects, which are not income of any factor of production.

5. Check the direction: Gross > Net; for most developing countries with foreign firms, GDP > GNP.

WORKED EXAMPLE 9 **From GDP to National Income**

In 2023, the economy of Ndoland recorded the following (billions of FCFA): GDP at market prices = 6 800; income received from abroad = 120; income paid to abroad = 420; depreciation = 500; indirect taxes = 800; subsidies = 150. Calculate: (a) GNP at market prices; (b) NNP at market prices; (c) National Income.

SOLUTION

(a) GNP at market prices. First compute net property income from abroad:

$$120 - 420 = -300$$

The negative value reflects profits repatriated by foreign companies (for example in the oil sector) exceeding what Ndolanders earn abroad. Then:

$$GNP_{mp} = 6\,800 + (-300) = 6\,500 \text{ billion FCFA}$$

(b) NNP at market prices. Subtract depreciation:

$$NNP_{mp} = 6\,500 - 500 = 6\,000 \text{ billion FCFA}$$

(c) National Income = NNP at factor cost. Subtract indirect taxes, add subsidies:

$$\begin{aligned} \text{NNP}_{fc} &= \text{NNP}_{mp} - \text{indirect taxes} + \text{subsidies} \\ &= 6\,000 - 800 + 150 \\ &= 5\,350 \text{ billion FCFA} \end{aligned}$$

Conclusion. National Income of Ndoland = 5 350 billion FCFA. Observe the logic of the chain: each step removes an element that is not income earned by the country's factors of production.

7.3 Skill 3: Computing National Income by the income approach

METHOD **Calculating National Income using the income approach**

1. Recall the principle: every franc spent on production becomes somebody's income. Add all incomes earned by factors of production within the country:

   $$\text{Domestic income} = W + R + i + P$$

   where W = wages and salaries (compensation of employees), R = rent, i = interest, P = profits (including undistributed profits and dividends).

2. Exclude transfer incomes: pensions, scholarships, gifts, unemployment benefits — they are received without any current production.

3. Add net property income from abroad to move from domestic income to national income.

4. Cross-check: the result should equal NNP_{fc} obtained by the other approaches (apart from statistical discrepancies).

Reflex: ask of every item: "Is this a payment for a factor of production (land, labour, capital, enterprise)?" If not, exclude it.

WORKED EXAMPLE 10 **National Income of Ndoland by the income approach**

From the records of Ndoland for 2023 (billions of FCFA): wages and salaries = 3 100; rents of land and buildings = 450; interest on capital = 600; profits of companies = 1 400; income of self-employed persons = 250; old-age pensions = 350; net property income from abroad = -300. Calculate the National Income.

SOLUTION

Step 1: Identify factor incomes.

- Wages and salaries: 3 100 (reward of labour) — include;
- Rents: 450 (reward of land) — include;
- Interest: 600 (reward of capital) — include;
- Profits: 1 400 (reward of enterprise) — include;
- Self-employed income: 250 (mixed labour/enterprise income) — include;
- Pensions: 350 — exclude (transfer payment, no service rendered in return).

Step 2: Sum domestic factor incomes.

$$\text{Domestic income} = 3\,100 + 450 + 600 + 1\,400 + 250$$
$$= 5\,800 \text{ billion FCFA}$$

Step 3: Add net property income from abroad.

$$\text{National Income} = 5\,800 + (-300) = 5\,500 \text{ billion FCFA}$$

Comment. The small difference with the 5 350 found by the expenditure chain is a statistical discrepancy, common in real accounts because data come from different sources. In an examination, if the question says the accounts are consistent, the two approaches must give the same figure — so always check your arithmetic.

7.4 Skill 4: Computing GDP by the output (value added) approach

METHOD **Calculating GDP using the output approach**

1. Avoid double counting: never add the total value of output of all firms, because one firm's output is another's input. Use value added instead:

$$\text{Value added} = \text{value of output} - \text{value of intermediate consumption}$$

2. Sum the value added of all sectors (primary, secondary, tertiary):

$$\text{GDP} = \sum \text{value added of all producing units}$$

3. Alternatively, count only the value of final goods (goods sold to the final user), which gives the same result.
4. Adjust if required: value added at basic prices + taxes on products − subsidies gives GDP at market prices.

Reflex: when a question gives a production chain (e.g. farmer → miller → baker), compute value added at each stage and sum.

WORKED EXAMPLE 11 **Value added in the cocoa chain**

In the Centre Region of Ndoland, the following transactions occur in a year (millions of FCFA): a farmer sells cocoa beans to a cooperative for 800; the cooperative dries, ferments and sells the beans to an exporter for 1300; the exporter sells them abroad for 1700. Assume no other intermediate inputs. (a) Calculate the value added at each stage. (b) Calculate the contribution of this chain to GDP. (c) Show that summing final output only gives the same answer.

SOLUTION

(a) Value added at each stage.

$$\text{Farmer: } 800 - 0 = 800 \quad \text{(no purchased inputs assumed)}$$
$$\text{Cooperative: } 1300 - 800 = 500$$
$$\text{Exporter: } 1700 - 1300 = 400$$

(b) Contribution to GDP.

$$\text{GDP} = 800 + 500 + 400 = 1700 \text{ million FCFA}$$

(c) Final output method. The only final output is the exported cocoa, worth 1700 million FCFA. The sales of 800 and 1300 are intermediate transactions: their value is already embodied in the final price. Summing final output gives 1700 million FCFA — identical. Key lesson. If we had naively added all sales ($800 + 1300 + 1700 = 3800$), we would have counted the farmer's cocoa three times. This is the **double counting** problem, and value added is the solution.

7.5 Skill 5: Nominal GDP, real GDP and the GDP deflator

METHOD **Converting nominal GDP into real GDP**

1. Distinguish the concepts: **nominal GDP** is measured at current prices; **real GDP** is measured at constant (base-year) prices and therefore reflects changes in quantities only.
2. Use the deflator formula:

$$\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{GDP deflator}} \times 100$$

3. Conversely:

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

4. Compute growth rates:

$$g = \frac{\text{Real GDP}_{t} - \text{Real GDP}_{t-1}}{\text{Real GDP}_{t-1}} \times 100$$

5. Interpret: if nominal GDP rises faster than prices, real output grows; if nominal GDP rises only because of inflation, real GDP is stagnant.

WORKED EXAMPLE 12 **Growth or inflation?**

Ndoland's nominal GDP was 6 800 billion FCFA in 2023 and 7 412 billion FCFA in 2024. The GDP deflator rose from 100 in 2023 to 109 in 2024. (a) Calculate real GDP for each year (base 2023). (b) Calculate the growth rate of real GDP. (c) Comment.

SOLUTION

(a) Real GDP.

$$\text{Real GDP}_{2023} = \frac{6\,800}{100} \times 100 = 6\,800 \text{ billion FCFA}$$

$$\text{Real GDP}_{2024} = \frac{7\,412}{109} \times 100 = 6\,800 \text{ billion FCFA}$$

(b) Growth rate.

$$g = \frac{6\,800 - 6\,800}{6\,800} \times 100 = 0\,\%$$

(c) Comment. Nominal GDP increased by

$$\frac{7\,412 - 6\,800}{6\,800} \times 100 = 9\,\%$$

but prices also rose by 9 % (deflator from 100 to 109). The entire increase in nominal GDP is therefore due to inflation: the physical volume of goods and services produced did not change at all. This is why economists — and GCE examiners — insist on using real GDP, not nominal GDP, to measure economic growth.

7.6 Skill 6: GDP per capita and international comparison

METHOD **Computing and interpreting per capita income**

1. Formula:
$$\text{GDP per capita} = \frac{\text{GDP (or GNP)}}{\text{total population}}$$
2. Growth of per capita income ≈ growth of GDP − growth of population. If population grows faster than GDP, per capita income falls even though total GDP rises.
3. Interpret with care: per capita income is an average; it hides inequality in distribution, the informal sector, and non-monetised subsistence production — all very significant in Cameroon and CEMAC countries.
4. For international comparisons, convert using exchange rates or, better, purchasing power parity (PPP), and use real figures.

WORKED EXAMPLE 13 **Per capita income in Ndoland**

In 2023, Ndoland's GDP was 6 800 billion FCFA and its population 20 million. In 2024, real GDP grew by 4 % while population grew by 2.6 %. (a) Calculate GDP per capita in 2023. (b) Estimate the growth rate of per capita income in 2024. (c) Give two reasons why this average may overstate the well-being of the typical citizen.

SOLUTION

(a) GDP per capita in 2023.

$$\frac{6\,800\,000\,000\,000}{20\,000\,000} = 340\,000 \text{ FCFA per person per year}$$

(b) Growth of per capita income.

$$g_{\text{per capita}} \approx 4\,\% - 2.6\,\% = 1.4\,\%$$

Average income rises, but much more slowly than total output, because the additional production must be shared among more people. This is a classic situation in fast-growing African economies. (c) Limitations of the average.

- Inequality: if oil revenues accrue to a small group while most farmers earn little, the average of 340 000 FCFA does not describe the typical citizen's life.
- Informal and subsistence activities: much food produced and consumed within rural households, petty trade and unrecorded services escape measurement, so the figure is imperfect in both directions.

Conclusion. Per capita GDP is a useful first indicator of development, but it must be complemented by data on distribution, health and education before judging welfare.

7.7 Skill 7: Verifying the equality of the three approaches

METHOD **Reconciling expenditure, income and output**

1. Remember the circular flow: every expenditure by a buyer is income to a seller, and every income arises from value added in production. Hence, in principle:

   Output approach = Income approach = Expenditure approach

2. In examinations, use this equality as a checking device: compute GDP by two approaches and confirm they match (or explain any stated "statistical discrepancy").

3. Common trap: mixing valuation bases. Ensure all three totals are at the same valuation (market prices or factor cost) before comparing; adjust with net indirect taxes if needed.

WORKED EXAMPLE 14 **A full reconciliation exercise**

The accounts of Ndoland (billions of FCFA) give: value added in agriculture 2 400; industry 1 900; services 2 300; taxes on products net of subsidies 200; wages 3 100; rent 450; interest 600; profits 1 400; self-employment income 250; consumption 4 200; investment 1 400; government spending 1 500; exports 1 800; imports 2 100; depreciation 500. (a) Compute GDP at market prices by the output approach. (b) Compute GDP at market prices by the income approach (add back depreciation and net indirect taxes to domestic income). (c) Compute GDP by the expenditure approach. (d) Conclude.

SOLUTION

(a) Output approach.

$$\text{GDP at basic prices} = 2\,400 + 1\,900 + 2\,300 = 6\,600$$
$$\text{GDP at market prices} = 6\,600 + 200 = 6\,800$$

(b) Income approach. Domestic factor income gives NNP-type measures at factor cost; to reach GDP at market prices we add depreciation and net indirect taxes:

$$\text{Domestic income} = 3\,100 + 450 + 600 + 1\,400 + 250 = 5\,800$$
$$+\text{depreciation} = 5\,800 + 500 = 6\,300$$
$$+\text{net indirect taxes} = 6\,300 + 200 = 6\,500$$

This gives 6 500, i.e. GNP at market prices; adding back net property income paid abroad of 300 (from Skill 2) yields 6 800. (c) Expenditure approach.

$$\text{GDP}_{mp} = 4\,200 + 1\,400 + 1\,500 + (1\,800 - 2\,100) = 6\,800$$

(d) Conclusion. All three approaches converge on 6 800 billion FCFA, confirming the fundamental identity of national income accounting: output = income = expenditure. In the GCE examination, presenting this reconciliation neatly, with each step labelled, earns full method marks even before the final figure.

WARNING **Frequent errors in national income calculations**

- Adding imports instead of subtracting them in the expenditure approach.
- Including transfer payments (pensions, scholarships) as income or government consumption.
- Double counting intermediate goods instead of using value added.
- Confusing gross and net: forgetting to deduct depreciation when asked for NNP or National Income.
- Mixing market prices and factor cost: always adjust for indirect taxes and subsidies when crossing from one valuation to the other.
- Using nominal GDP to measure growth without deflating for inflation.

8 EXERCISES

8.1 I check my knowledge

EXERCISE 5 **MCQ: National Income Concepts**

Choose the correct answer for each question.

1. Which of the following is included in GDP at market prices but excluded from GDP at factor cost?
   a) Consumption of fixed capital
   b) Indirect taxes less subsidies
   c) Net property income from abroad
   d) Compensation of employees
2. Gross National Product (GNP) equals GDP plus:
   a) Net exports
   b) Net property income from abroad
   c) Depreciation
   d) Indirect taxes
3. In the expenditure approach, which component represents spending on new capital goods?
   a) Consumption (C)
   b) Government spending (G)
   c) Investment (I)
   d) Net exports (X–M)
4. Real GDP is obtained by:
   a) Dividing nominal GDP by the GDP deflator and multiplying by 100
   b) Multiplying nominal GDP by the consumer price index
   c) Adding depreciation to nominal GDP
   d) Subtracting indirect taxes from nominal GDP

EXERCISE 6 **True/False with Justification**

State whether each statement is true or false and provide a brief justification.

1. Transfer payments such as pensions are included in national income because they represent income to households.
2. The value of intermediate goods is included in GDP to avoid double counting.
3. Net factor income from abroad is added to GDP to obtain GNP.
4. The output approach sums the value of sales of all firms in the economy.

EXERCISE 7 **Definitions**

Define the following terms precisely:

1. Gross Domestic Product at market prices (GDP_{mp})
2. Net National Product at factor cost (NNP_{fc}) — also called National Income
3. Gross Domestic Fixed Capital Formation
4. Statistical discrepancy

EXERCISE 8 **Direct Calculations: GDP and GNP**

Using the data below for a hypothetical economy (in billion CFA francs), calculate:

1. GDP at market prices (expenditure approach)
2. GNP at market prices
3. National Income (NNP at factor cost)

Item	Value (billion CFA)
Private final consumption expenditure (C)	12,000
Government final consumption expenditure (G)	3,500
Gross domestic fixed capital formation (I)	4,200
Change in stocks	300
Exports of goods and services (X)	2,800
Imports of goods and services (M)	3,100
Consumption of fixed capital (Depreciation)	1,100
Indirect taxes	1,500
Subsidies	400
Net property income from abroad	-200

8.2 I practise

EXERCISE 9 **Expenditure Approach Calculation**

The table below shows the expenditure components for Cameroon in a given year (in billion CFA francs).

Component	Value
Household final consumption expenditure	18,500
NPISH final consumption expenditure	400
General government final consumption expenditure	4,200
Gross fixed capital formation	5,600
Changes in inventories	200
Exports of goods and services	3,800
Imports of goods and services	4,100

1. Calculate GDP at market prices using the expenditure approach.
2. If consumption of fixed capital is 1,300, indirect taxes are 1,800 and subsidies are 500, compute GDP at factor cost.
3. Briefly explain why changes in inventories are treated as part of investment.

EXERCISE 10 **Income Approach Calculation**

The following factor incomes and other items are recorded for an economy (in billion CFA francs).

Item	Value
Compensation of employees	14,000
Operating surplus (rent, interest, profit)	9,500
Mixed income (self-employed)	3,200
Consumption of fixed capital	1,100
Indirect taxes	1,600
Subsidies	300
Net factor income from abroad	-150
Transfer payments (pensions, scholarships)	800

1. Calculate GDP at factor cost using the income approach.
2. Calculate GNP at market prices.
3. Explain why transfer payments are excluded from national income.

EXERCISE 11 **Output (Value Added) Approach**

Consider a simple production chain in the maize sector:

- A farmer grows maize and sells it to a miller for 500,000 CFA francs.
- The miller processes the maize into flour and sells it to a baker for 800,000 CFA francs.
- The baker bakes bread and sells it to consumers for 1,200,000 CFA francs.

Assume no other inputs are purchased from outside this chain.

1. Compute the value added at each stage.
2. Show that the sum of value added equals the final market value of the bread.
3. Explain why summing the sales values (500,000 + 800,000 + 1,200,000) would overstate total output.

EXERCISE 12 **Nominal vs Real GDP Conversion**

The table gives nominal GDP and a GDP deflator (base year = 2015, index = 100) for three years.

Year	Nominal GDP (billion CFA)	GDP Deflator
2015	25,000	100
2018	30,000	115
2021	36,000	130

1. Compute real GDP for each year (base year 2015).
2. Calculate the annual average growth rate of real GDP between 2015 and 2021.
3. Comment on the difference between nominal and real GDP growth over this period.

8.3 I prepare for the GCE

EXERCISE 13 **Essay: Problems of Measuring National Income in Cameroon**

Question: Discuss the major problems encountered in measuring national income in Cameroon and suggest possible solutions. (25 marks)

EXERCISE 14 **Essay: Uses and Limitations of National Income Statistics**

Question: Explain five uses of national income statistics to the government of Cameroon and three limitations of these statistics as a measure of welfare. (25 marks)

EXERCISE 15 **Data Response: Comprehensive National Income Accounting**

The following data relate to a hypothetical economy for the year 2023 (in billion CFA francs).

Item	Value
Private consumption expenditure	20,000
Government consumption expenditure	5,000
Gross fixed capital formation	6,000
Change in stocks	400
Exports	4,500
Imports	5,200
Compensation of employees	15,000
Operating surplus	10,000
Mixed income	3,500
Consumption of fixed capital	1,200
Indirect taxes	2,000
Subsidies	600
Net factor income from abroad	-300
Transfer payments	1,000

1. Calculate GDP at market prices using the expenditure approach. (4 marks)
2. Calculate GDP at factor cost using the income approach. (4 marks)
3. Derive GNP at market prices and National Income (NNP at factor cost). (4 marks)
4. Explain the treatment of each of the following in national income accounting: (a) change in stocks, (b) subsidies, (c) transfer payments. (6 marks)
5. Identify two possible sources of statistical discrepancy between the expenditure and income estimates. (3 marks)
6. Briefly discuss why GDP per capita may not be a good indicator of the standard of living in this economy. (4 marks)

9 SOLUTIONS TO THE EXERCISES

9.1 Solutions — I check my knowledge

SOLUTION **Exercise 1 — MCQ: National Income Concepts**

Question 1: answer b) — **Indirect taxes less subsidies**.
GDP at market prices (GDP_{mp}) values output at the prices actually paid by buyers, which include **indirect taxes** (such as VAT) and are reduced by **subsidies**. To move from market prices to **factor cost** we subtract net indirect taxes:

$$\text{GDP}_{fc} = \text{GDP}_{mp} - \text{indirect taxes} + \text{subsidies}.$$

Hence indirect taxes less subsidies are in GDP_{mp} but not in GDP_{fc}. **Consumption of fixed capital** distinguishes gross from net aggregates, **net property income from abroad** distinguishes domestic from national aggregates, and **compensation of employees** is a factor income present in both.
Question 2: answer b) — **Net property income from abroad**.
GNP measures the income of residents, wherever it is earned, while **GDP** measures production within the territory. The bridge is:

$$\text{GNP} = \text{GDP} + \text{net property (factor) income from abroad}.$$

Question 3: answer c) — **Investment (I)**.
Investment (gross capital formation) is spending on new capital goods: machinery, buildings, infrastructure, plus **changes in stocks**. Consumption is household spending, G is government current spending, and $(X - M)$ is the **foreign balance**.
Question 4: answer a).
Real GDP removes the effect of price changes:

$$\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{GDP deflator}} \times 100.$$

SOLUTION **Exercise 2 — True/False with Justification**

1. FALSE. **Transfer payments** (pensions, scholarships, unemployment benefits) are not included in national income because they are not payments for any current production of goods or services; they are mere redistributions of existing income. Including them would double count income already earned by the taxpayers who financed them.
2. FALSE. It is precisely to avoid **double counting** that intermediate goods are excluded

from GDP. Their value is already embodied in the value of the final goods they helped to produce. Only final goods and services (or, equivalently, **value added** at each stage) enter GDP.
3. TRUE. By definition GNP = GDP + net factor income from abroad. This adjustment converts a domestic (territorial) concept into a national (residents') concept. Note the adjustment can be negative, as is typical for Cameroon where profit repatriation by foreign firms exceeds income received from abroad.
4. FALSE. The **output approach** sums the value added of each firm (sales value minus the cost of intermediate inputs purchased from other firms), not the gross value of sales. Summing sales values would count intermediate output several times and grossly overstate production.

SOLUTION **Exercise 3 — Definitions**

1. **GDP at market prices** (GDP_{mp}): the total market value of all final goods and services produced within the geographical territory of a country during a given period (usually one year), valued at the prices paid by purchasers, i.e. inclusive of indirect taxes and net of subsidies.
2. **NNP at factor cost** (NNP_{fc}), or **National Income**: the total income earned by the residents of a country from their factors of production (labour, land, capital, entrepreneurship) during a year, after deducting **depreciation** (consumption of fixed capital):

$$NNP_{fc} = GNP_{mp} - \text{depreciation} - \text{indirect taxes} + \text{subsidies}.$$

3. **Gross Domestic Fixed Capital Formation** (GDFCF): the value of expenditure on new fixed capital assets (machinery, vehicles, buildings, roads, other construction) produced within the country during the period, before deducting consumption of fixed capital. It is "gross" because it includes replacement investment.
4. **Statistical discrepancy**: the residual balancing item inserted in national accounts so that the expenditure, income and output estimates of GDP coincide. It arises because the three approaches use different data sources which never match perfectly due to errors, omissions, informal activity and timing differences.

SOLUTION **Exercise 4 — Direct Calculations: GDP and GNP**

1. GDP at market prices (expenditure approach). Investment includes both fixed capital formation and the change in stocks: $I = 4{,}200 + 300 = 4{,}500$.

$$\begin{aligned} GDP_{mp} &= C + I + G + (X - M) \\ &= 12{,}000 + 4{,}500 + 3{,}500 + (2{,}800 - 3{,}100) \\ &= \boxed{19{,}700 \text{ billion CFA}}. \end{aligned}$$

2. GNP at market prices.

GNP_{mp} = GDP_{mp} + net property income from abroad = 19,700 + (−200) = **19,500 billion CFA**.

3. National Income (NNP_{fc}). Two steps: remove depreciation (gross → net), then remove net indirect taxes (market prices → factor cost).

$$NNP_{mp} = 19{,}500 - 1{,}100 = 18{,}400,$$
$$NNP_{fc} = 18{,}400 - (1{,}500 - 400) = 18{,}400 - 1{,}100 = \boxed{17{,}300 \text{ billion CFA}}.$$

Check: net indirect taxes = 1,500 − 400 = 1,100; coincidentally equal to depreciation here — do not confuse the two items.

9.2 Solutions — I practise

SOLUTION **Exercise 5 — Expenditure Approach Calculation**

1. GDP at market prices. Private consumption C includes households and NPISH: C = 18,500 + 400 = 18,900. Investment I = 5,600 + 200 = 5,800.

$$GDP_{mp} = C + I + G + (X - M)$$
$$= 18{,}900 + 5{,}800 + 4{,}200 + (3{,}800 - 4{,}100)$$
$$= 28{,}900 - 300 = \boxed{28{,}600 \text{ billion CFA}}.$$

2. GDP at factor cost.

GDP_{fc} = GDP_{mp} − indirect taxes + subsidies = 28,600 − 1,800 + 500 = **27,300 billion CFA**.

(Consumption of fixed capital is not needed here: it would be used to move from gross to net domestic product, NDP_{fc} = 27,300 − 1,300 = 26,000.)

3. Why changes in inventories are investment. National income accounting requires that everything produced during the year be matched by an expenditure. Unsold goods produced this year (an increase in stocks) have no buyer yet, so they are conventionally treated as if the firm "purchased" them itself — a form of **investment**. Conversely, a fall in stocks means part of current sales came from past production and must be subtracted. This keeps production and expenditure equal.

SOLUTION **Exercise 6 — Income Approach Calculation**

1. GDP at factor cost. Sum all factor incomes earned within the territory (transfer pay-

ments excluded):

$$\text{GDP}_{fc} = \text{compensation of employees} + \text{operating surplus} + \text{mixed income}$$
$$= 14{,}000 + 9{,}500 + 3{,}200 = \boxed{26{,}700 \text{ billion CFA}}.$$

2. GNP at market prices. First convert factor cost to market prices, then add net factor income from abroad:

$$\text{GDP}_{mp} = \text{GDP}_{fc} + \text{indirect taxes} - \text{subsidies} = 26{,}700 + 1{,}600 - 300 = 28{,}000,$$
$$\text{GNP}_{mp} = \text{GDP}_{mp} + \text{net property income from abroad} = 28{,}000 + (-150)$$
$$= \boxed{27{,}850 \text{ billion CFA}}.$$

(Depreciation would only be needed to compute net aggregates: $\text{NNP}_{fc} = 26{,}700 - 1{,}100 - 150 = 25{,}450$.)

3. Why transfers are excluded. **Transfer payments** (pensions, scholarships, family allowances) are not made in exchange for any current factor service or output. They redistribute income already created and counted elsewhere. Adding them to factor incomes would therefore double count and overstate national income.

SOLUTION **Exercise 7 — Output (Value Added) Approach**

1. Value added at each stage. Value added = sales value − value of intermediate inputs bought from other firms.

Stage	Sales (CFA)	Intermediate inputs (CFA)	Value added (CFA)
Farmer (maize)	500,000	0	500,000
Miller (flour)	800,000	500,000	300,000
Baker (bread)	1,200,000	800,000	400,000
Total	2,500,000	1,300,000	1,200,000

2. Equality with final value. Sum of value added = 500,000 + 300,000 + 400,000 = 1,200,000 CFA, exactly the final market value of the bread. This illustrates the fundamental identity: sum of value added = value of final output = sum of factor incomes generated.

3. Why summing sales overstates output. Summing sales (500,000 + 800,000 + 1,200,000 = 2,500,000) counts the maize three times and the flour twice: the value of intermediate goods is embedded in the price of the goods made from them. This **double counting** would overstate total output by 1,300,000 CFA. Only value added — or the value of final goods alone — gives a correct measure of production.

SOLUTION **Exercise 8 — Nominal vs Real GDP Conversion**

1. Real GDP (base 2015). Using Real GDP = $\frac{\text{Nominal GDP}}{\text{deflator}} \times 100$:

Year	Nominal GDP	Deflator	Real GDP (billion CFA)
2015	25,000	100	25,000 × 100/100 = 25,000
2018	30,000	115	30,000 × 100/115 ≈ 26,087
2021	36,000	130	36,000 × 100/130 ≈ 27,692

2. Average annual growth rate of real GDP, 2015–2021. Over 6 years, using the compound (geometric) rate:

$$g = \left[\left(\frac{27{,}692}{25{,}000}\right)^{1/6} - 1\right] \times 100 \approx \left[(1.1077)^{1/6} - 1\right] \times 100 \approx 1.71\,\% \text{ per year.}$$

(A simple average approximation, $\frac{27{,}692-25{,}000}{25{,}000} \div 6 \approx 1.79\,\%$, is acceptable if the method is stated.)

3. Comment. Nominal GDP grew by $\frac{36{,}000-25{,}000}{25{,}000} = 44\,\%$ over the period, but real GDP grew by only about 10.8 %. The difference — roughly 30 % cumulative, matching the rise of the deflator from 100 to 130 — reflects pure price increases (inflation). Nominal growth therefore overstates the true expansion of production; only real GDP measures the change in the volume of goods and services available.

9.3 Solutions — I prepare for the GCE

SOLUTION **Exercise 9 — Essay: Problems of Measuring National Income in Cameroon**

Examiner expectations (25 marks): a clear introduction defining national income; at least five well-developed problems, each with a Cameroonian illustration; matched solutions; a brief evaluative conclusion. Indicative scheme: introduction 2 marks; each problem + solution about 4 marks (up to 20); conclusion 3 marks.

Introduction. National income is the money value of all final goods and services produced by a country in a year. In Cameroon, as in most developing economies, statisticians face serious difficulties in measuring it accurately.

Problems and solutions.

1. Large informal (subsistence) sector. Much output — subsistence farming, petty trade, street vending, artisan work — is never recorded or passes through no market. Solution: regular household and informal-sector surveys, and imputation of the value of own-consumed output.

2. Inadequate and unreliable data. Many small firms keep no proper accounts; respondents give false figures for fear of taxation. Solution: strengthen the national statis-

tics institute, guarantee confidentiality, and separate statistical collection from tax administration.

3. Double counting. Distinguishing final from intermediate goods is difficult (e.g. cassava sold or processed at home). Solution: systematic use of the value-added method and input-output tables.

4. Valuation of non-marketed services. Owner-occupied housing, unpaid family labour and government services (defence, administration) have no market price. Solution: impute rents and value public services at cost, following international conventions (SNA).

5. Shortage of skilled personnel and funds. Few trained statisticians and limited budgets delay and weaken data collection. Solution: train statisticians, computerise records, and cooperate with international bodies (IMF, World Bank, AFRISTAT).

6. Price instability and the underground economy. Inflation distorts nominal values, while illegal or hidden activities (smuggling across porous borders) escape measurement. Solution: compute real aggregates with price deflators and estimate hidden activity through surveys.

Conclusion. Because of these problems, Cameroon's national income figures are estimates with a margin of error; improving statistical capacity and formalising the informal sector would make them more reliable.

SOLUTION **Exercise 10 — Essay: Uses and Limitations of National Income Statistics**

Examiner expectations (25 marks): five uses clearly explained (about 3 marks each = 15), three welfare limitations (about 3 marks each = 9), introduction/conclusion 1 mark.
Five uses to the government of Cameroon.

1. Economic planning and policy formulation: national income data reveal the size and structure of the economy (agriculture, oil, services), guiding development plans, budgets and sectoral policies.

2. Measuring economic growth and performance: comparing real GDP over time shows whether the economy is growing, stagnating or in recession, and whether policies succeed.

3. International comparison and aid allocation: GDP per capita allows comparison with other countries and is used by the IMF, World Bank and donors to classify countries and allocate loans and aid.

4. Assessing living standards and distribution: per capita income and its sectoral/regional breakdown help identify poverty pockets and design redistribution policies.

5. Taxation and resource mobilisation: income statistics show the tax base and the capacity of sectors to bear taxes, informing fiscal policy and CEMAC budgetary surveillance.

Three limitations as a measure of welfare.

1. Ignores income distribution: a high GDP per capita may coexist with mass poverty if oil or cocoa income is concentrated in few hands.
2. Excludes non-market and informal production: subsistence food, household work and leisure — important for welfare in rural Cameroon — are omitted or undervalued.
3. Ignores negative externalities and quality of life: pollution from extractive industries, deforestation, congestion, crime and stress reduce welfare but may raise GDP; GDP also says nothing about health, education or freedom.

Conclusion. National income statistics are indispensable for economic management, but GDP per capita is at best a partial proxy for welfare and should be complemented by indicators such as the Human Development Index.

SOLUTION **Exercise 11 — Data Response: Comprehensive National Income Accounting**

1. GDP at market prices (expenditure approach) — 4 marks.

$$I = 6{,}000 + 400 = 6{,}400; \quad X - M = 4{,}500 - 5{,}200 = -700.$$

$$\text{GDP}_{mp} = 20{,}000 + 6{,}400 + 5{,}000 - 700 = \boxed{30{,}700 \text{ billion CFA}}.$$

Mark scheme: correct formula (1); inclusion of change in stocks in I (1); correct net exports (1); correct result (1).

2. GDP at factor cost (income approach) — 4 marks.

$$\text{GDP}_{fc} = 15{,}000 + 10{,}000 + 3{,}500 = \boxed{28{,}500 \text{ billion CFA}}.$$

Mark scheme: correct identification of the three factor incomes (2); exclusion of transfer payments (1); correct result (1). Check of consistency: $\text{GDP}_{fc} = 30{,}700 - 2{,}000 + 600 = 29{,}300$; the gap of 800 with 28,500 is the statistical discrepancy (see part 5).

3. GNP at market prices and National Income — 4 marks.

$$\text{GNP}_{mp} = 30{,}700 + (-300) = \boxed{30{,}400 \text{ billion CFA}},$$

$$\text{NNP}_{fc} = \text{GNP}_{mp} - \text{depreciation} - \text{indirect taxes} + \text{subsidies}$$

$$= 30{,}400 - 1{,}200 - 2{,}000 + 600 = \boxed{27{,}800 \text{ billion CFA}}.$$

Mark scheme: 1 mark per correct step (net factor income adjustment, depreciation, net indirect taxes, result).

4. Treatment of items — 6 marks (2 each).

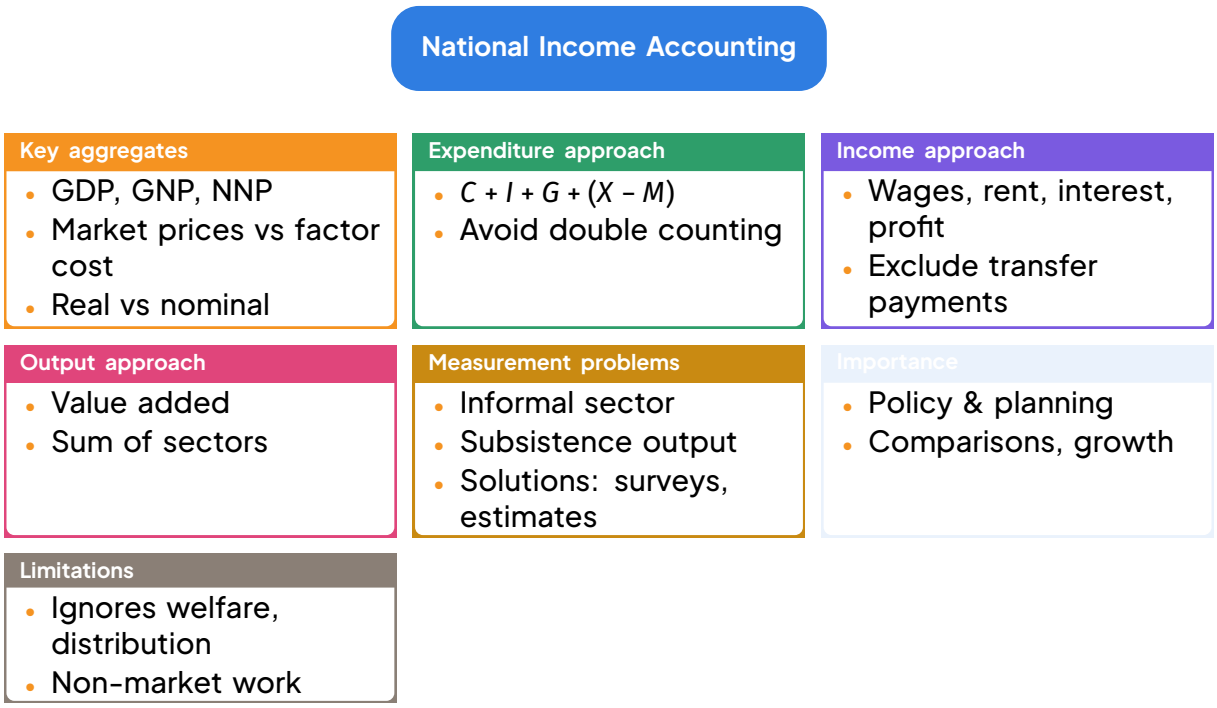
- (a) Change in stocks: included in investment. An increase in stocks is output produced but unsold during the year; it is treated as if firms bought it themselves, so that production equals expenditure. A decrease is subtracted.
- (b) Subsidies: subtracted when moving from market prices to factor cost (added when moving from factor cost to market prices). Subsidies lower market prices below factor cost, so they must be removed to obtain the income actually received by factors.
- (c) Transfer payments: excluded entirely from national income. They are not rewards for current production but redistributions (pensions, scholarships); including them would double count income.

5. Sources of statistical discrepancy — 3 marks. Any two, briefly explained:

- unrecorded informal-sector output and incomes, captured differently by the two approaches;
- errors, omissions and under-declaration in tax and survey returns (e.g. incomes hidden to evade tax);
- timing and valuation differences between expenditure records and income records.

6. GDP per capita and standard of living — 4 marks. Any two developed points: GDP per capita ignores the distribution of income (average may hide wide inequality); it omits informal and subsistence production important to household welfare; it takes no account of working hours, health, education, pollution or other non-material aspects of well-being; and with negative net factor income from abroad, part of domestic production accrues to foreigners, so residents' actual income is lower than GDP suggests. Hence GDP per capita is an imperfect indicator of the standard of living.

10 SUMMARY SHEET



Mind map of the chapter: National income accounting

FICHE BILAN **L'essentiel à connaître pour l'examen**

1. Key definitions

- **National income**: the total money value of all final goods and services produced by the residents of a country in a given period (usually one year), measured net of depreciation.
- **GDP** (Gross Domestic Product): value of final output produced within the country's borders, whoever produces it.
- **GNP** (Gross National Product): GDP + **net property income from abroad** (incomes earned by residents abroad minus incomes earned by foreigners at home).
- **NNP** = GNP − **depreciation** (capital consumption).
- **Market prices** vs **factor cost**: National income at factor cost = value at market prices − indirect taxes + subsidies.
- **Nominal** (current-price) income vs **real** (constant-price) income: real income removes the effect of inflation.

- **Per capita income** = national income ÷ population.
- **Transfer payments** (pensions, unemployment benefits, gifts): incomes received without any current production; excluded from national income.
- **Value added**: value of output − value of intermediate inputs bought from other firms.

2. Formulas to know by heart

- Expenditure approach:

$$GDP = C + I + G + (X - M)$$

  where C = household consumption, I = gross investment, G = government current spending on goods and services, X = exports, M = imports.
- Income approach: National income = wages and salaries + rent + interest + profits (+ mixed incomes of the self-employed), all earned by residents.
- Output approach: GDP = sum of **value added** of all sectors (primary, secondary, tertiary).
- GNP = GDP + net property income from abroad; NNP = GNP − depreciation; NI (factor cost) = NNP at market prices − indirect taxes + subsidies.

3. Methods (practical work)

- Expenditure method: (i) list final expenditures; (ii) add $C + I + G$; (iii) add exports and subtract imports; (iv) adjust to factor cost or to GNP/NNP if required.
- Income method: (i) select only factor incomes; (ii) exclude transfer payments and capital gains; (iii) sum; (iv) adjust for net property income and depreciation.
- Output method: (i) compute value added at each stage; (ii) sum over all firms/sectors; (iii) never sum gross output values (double counting).
- The three approaches give the same total in principle, because every franc spent is a franc earned and a franc of value added.

4. Problems of measurement and solutions (Cameroon context)

- Large **informal sector** and **subsistence agriculture** (family food farms) escape records → solution: sample surveys, imputed values.
- Poor statistical records, illiteracy, dishonest tax declarations → strengthen statistical services (INS), tax incentives for declaration.
- Valuing unpaid services (housewives' work), barter, and output consumed by producers → imputation at market prices.
- Price changes distort comparisons → use constant prices (real income) and price indices.
- Double counting → count only final goods or value added.

5. Importance of national income statistics

- Measure **economic growth** and compare living standards over time and between countries.
- Guide government **planning**, budgeting and policy (taxation, investment, AfCFTA strategy).
- Show the **structure** of the economy (share of agriculture, industry, services) and sectoral contributions.
- Basis for **international comparisons**, foreign aid allocation and debt assessment.
- Help firms assess market size and investors take decisions.

6. Limitations

- Says nothing about income **distribution**, leisure, welfare, or environmental damage.
- Excludes non-market and illegal activities; averages hide regional inequalities.

7. Common mistakes and GCE tips

- Mistake: including transfer payments or second-hand sales in the income/expenditure totals. Tip: ask "was anything newly produced?".
- Mistake: adding imports instead of subtracting them. Tip: imports are produced abroad.
- Mistake: confusing GDP and GNP. Tip: GDP = territory; GNP = residents.
- Mistake: summing gross output in the output approach. Tip: use value added only.
- GCE tip: in calculation questions, show each step, state the formula first, give units (millions of FCFA), and end with a one-line interpretation; in essays, always evaluate (importance and limitations).

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